

STRUCTURAL MODULE 2

STRUCTURAL DESIGN WITH MASS TIMBER

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STRUCTURAL MODULE 2: Structural Design with Mass Timber

Building on [Structural Module 1: Structural Basics of Mass Timber](#), this module outlines key factors to consider when approaching the design of a mass timber building. Efficient mass timber structural design benefits from an integrated, interdisciplinary process in which early decisions align design intent, code compliance, material performance, systems integration, and practical constraints, including cost and constructibility.

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Laura Britton is an architect, author, and educator with a focus on environmentally- and socially-conscious design. As a Senior Associate at Shigeru Ban Architects, she has led and contributed to diverse projects ranging from complex mass timber structures to humanitarian relief efforts. Britton is the Co-Author and Editor of Shigeru Ban: Timber in Architecture, a book tracing the evolution of 45 wood projects from concept through construction. She is an Adjunct Assistant Professor at The Cooper Union and Co-Founder of Mass Timber Pedagogy.

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Mass Timber Structural Design Process

As a mass timber building is designed and developed, consider the following:

BASIC DESIGN CRITERIA

- Motivations: Why mass timber? Why not mass timber?
- Code: Construction type, fire-resistance rating (FRR), and allowable height and area

DESIGN FOR PERFORMANCE

- Acoustics
- Mechanical, electrical, and plumbing (MEP)

DESIGN FOR ECONOMY

- Unit cost and volume of timber
- Optimizing the structural grid
- Design for manufacturing and assembly (DfMA)

DESIGN FOR CONSTRUCTIBILITY

- Supply chain
- Manufacturing standards
- Connection design
- Material tolerances

This module is organized to sequentially address the topics outlined above.



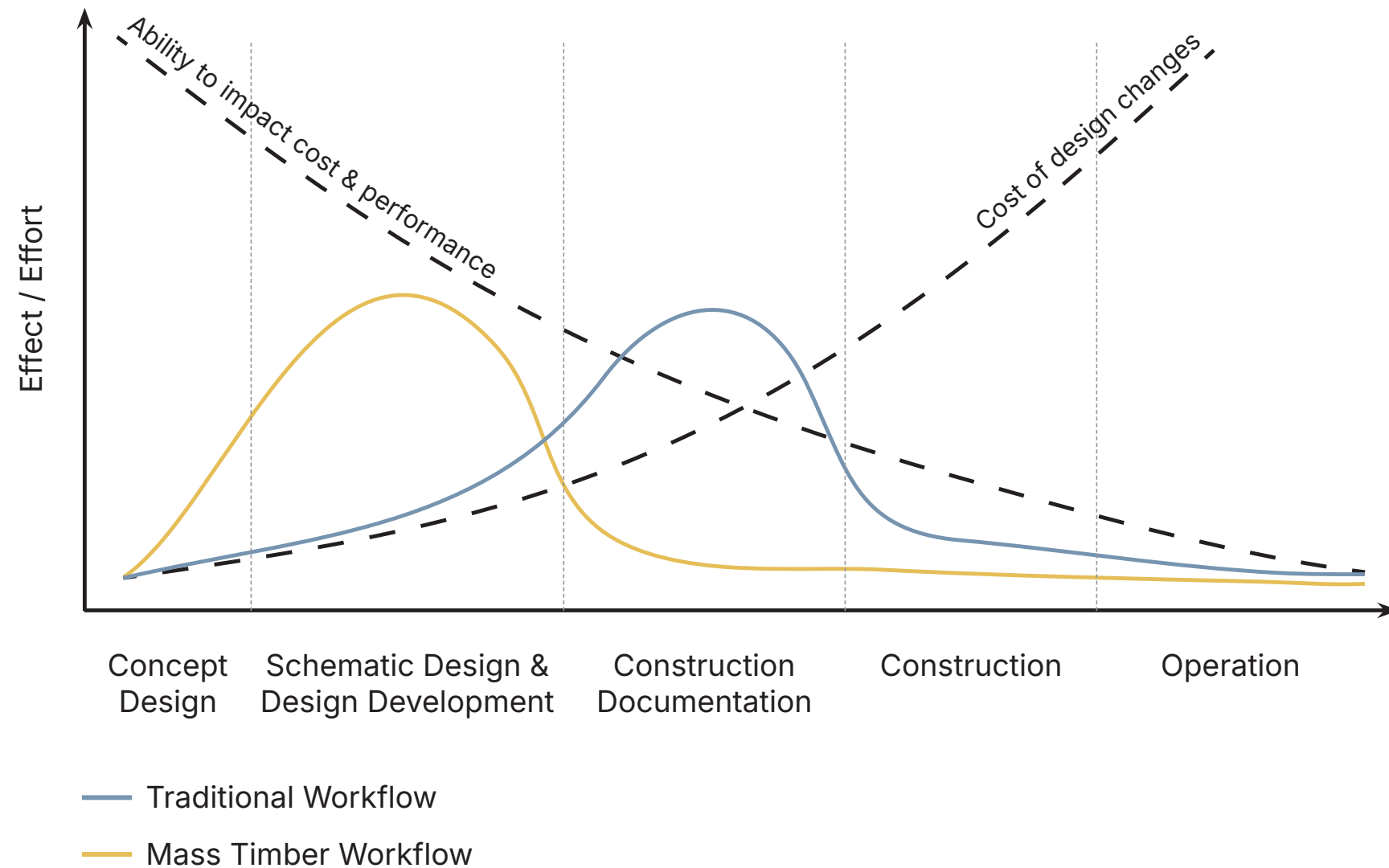
Nine Bridges Golf Clubhouse, Yeoju, Korea, 2010; Architect: Shigeru Ban Architects; Structural Engineer: Création Holz

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Mass Timber Structural Design Process

MACLEAMY CURVE, ADAPTED FOR MASS TIMBER WORKFLOW



The MacLeamy Curve, developed by Patrick MacLeamy, FAIA, illustrates how a project team's ability to influence design outcomes decreases as a project progresses, while the cost of making changes increases. In a "traditional" workflow, changes are most easily made during early project phases, but become increasingly limited in scope and costly in later phases. MacLeamy advocates a more integrated design process, shifting interdisciplinary collaboration to earlier project phases to both maximize impact and minimize cost.

For this graph, MacLeamy's curve for early design effort in Integrated Project Delivery is renamed "Mass Timber Workflow." The same principles apply, but to a greater extent: interdisciplinary factors should be considered early in a project.

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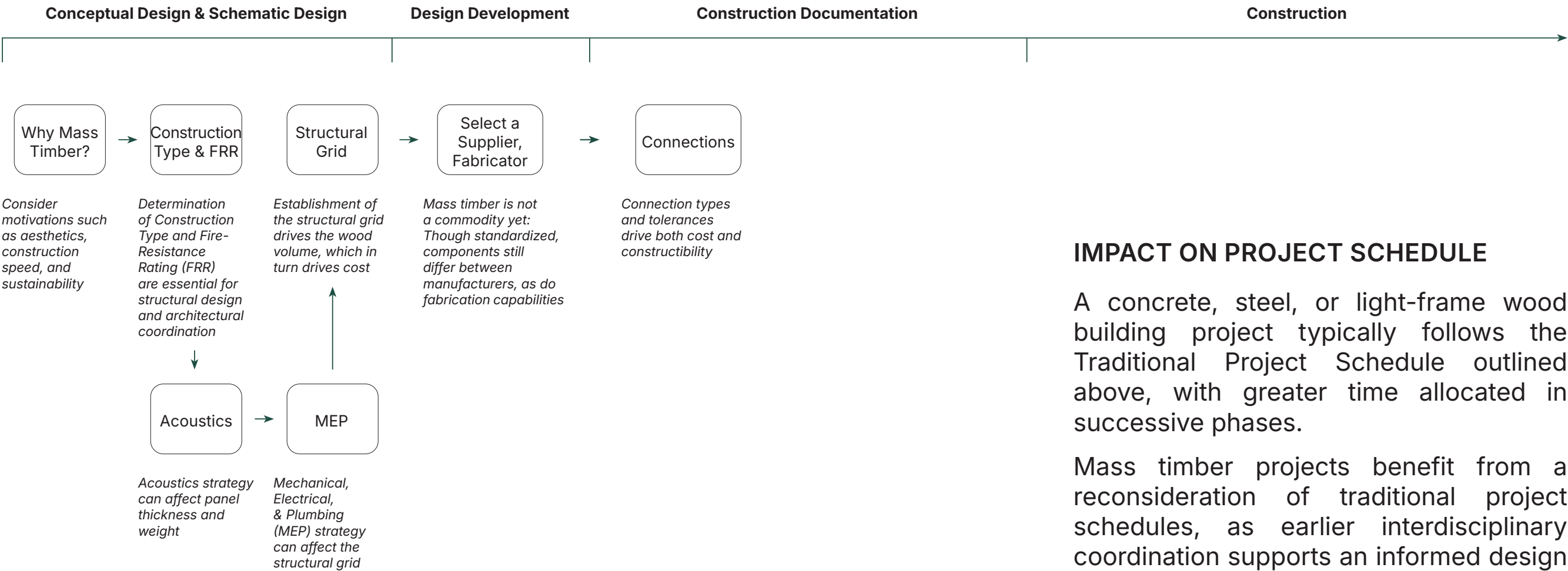
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Mass Timber Structural Design Process

TRADITIONAL PROJECT SCHEDULE



MASS TIMBER PROJECT SCHEDULE



IMPACT ON PROJECT SCHEDULE

A concrete, steel, or light-frame wood building project typically follows the Traditional Project Schedule outlined above, with greater time allocated in successive phases.

Mass timber projects benefit from a reconsideration of traditional project schedules, as earlier interdisciplinary coordination supports an informed design process with fabrication, procurement, and cost control in mind. This logic aligns with the values espoused in MacLeamy’s Curve: design impact is greatest, and changes are least costly, when critical decisions are made early.

Basic Design Criteria: Why Mass Timber?

As mass timber increases in popularity, there is a growing consensus that “wood is good.” But mass timber is not a one-size-fits-all solution: it is important to consider the motivations for working with this family of materials, and how decisions early in the design and construction process can help meet project goals.

Some of the many appealing reasons to consider mass timber include:

ENVIRONMENTAL PERFORMANCE

Forest Stewardship

The use of responsibly sourced materials supports:

- Responsible forest management
- Certified supply chains
- Fire-risk mitigation in forests
- Watershed health
- Carbon sequestration

Responsible Material Selection: Wood

- Natural, renewable material
- Low embodied carbon
- Carbon storage: approximately 50% by dry weight
- Potential for disassembly and reuse
- Supports rural and regional manufacturing economies

MATERIAL PERFORMANCE

- Composed of wood from small diameter trees rather than old growth, benefiting forest health and fire management
- The use of multiple connected pieces of wood diminishes the impact of individual knots and flaws



Sawmill Facility, Kalesnikoff, British Columbia, Canada

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Basic Design Criteria: Why Mass Timber?

CONSTRUCTION

- Accelerated construction schedules, averaging 25% faster on-site construction¹
- Reduced on-site labor, with up to 70% fewer workers
- Improved job site conditions
 - Quieter
 - Cleaner
 - Reduced disruption
- Material efficiency due to prefabrication
- Reduced construction traffic by up to 90%

FINISHED BUILDINGS

Architecture

- Design intent and aesthetic appeal
- Material legibility
- Exposed structure and tectonic clarity
- Occupant experience — biophilia
- Compatibility with other material and structural systems, such as concrete and steel

Market Outcomes

- High lease rates
- High leasing velocity, often before completion of construction



Construction photo of Mercer Middle School



Mercer Middle School, Seattle, WA, 2025; Architect: Bassetti Architects, Structural Engineer: Coughlin Porter Lundeen

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Basic Design Criteria: Why Not Mass Timber?

Though mass timber projects have numerous benefits, there are circumstances when mass timber may not be the ideal material system. However, these potential pitfalls can be opportunities for innovation. For example:

POTENTIAL DETERRENTS

- The team (architects, engineers, and builders) is inexperienced with mass timber
- Project delivery is Design / Bid / Build, with little opportunity to engage with the expertise of manufacturers in a timely manner
- Wood is exposed to the exterior, or conditions demand durability in wet/dry conditions, like parking garages
- Long spans with shallow structural depth
- Building is outside code limitations of height, stories, and area for mass timber
- Occupancy demands exceptional acoustic separation
- Occupancy demands exceptional vibration performance, such as laboratories with high-performance optical equipment

ALTERNATIVE CONSIDERATIONS

- If the goal of using mass timber is known from the outset of a project, select a project team with experience working with the material
- Consult experts. For example, WoodWorks provides free project support
- Mass timber may not show well with low-cost focus by trade. It is not a commodity
- Dry wood can last for centuries. Consider detailing strategies and weather protections
- Consider program needs and requisite structural depths early in the design process to mitigate, and whether other materials, or hybrid solutions with mass timber in combination with other materials, are better suited to project-specific demands
- Consider selective use of mass timber, such as a feature space or lobby
- Determine whether there is (literal) space to accommodate acoustical treatment in walls and floor assemblies, or use mass timber selectively
- Consider techniques to mitigate vibration, or hybrid solutions with mass timber in combination with other materials

Basic Design Criteria: Construction Type & Fire-Resistance Rating

THE INTERNATIONAL BUILDING CODE (IBC)

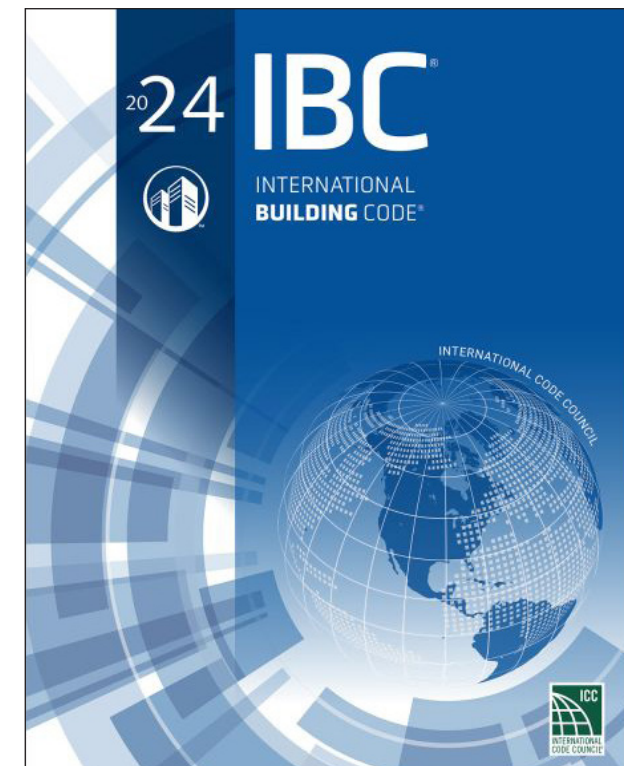
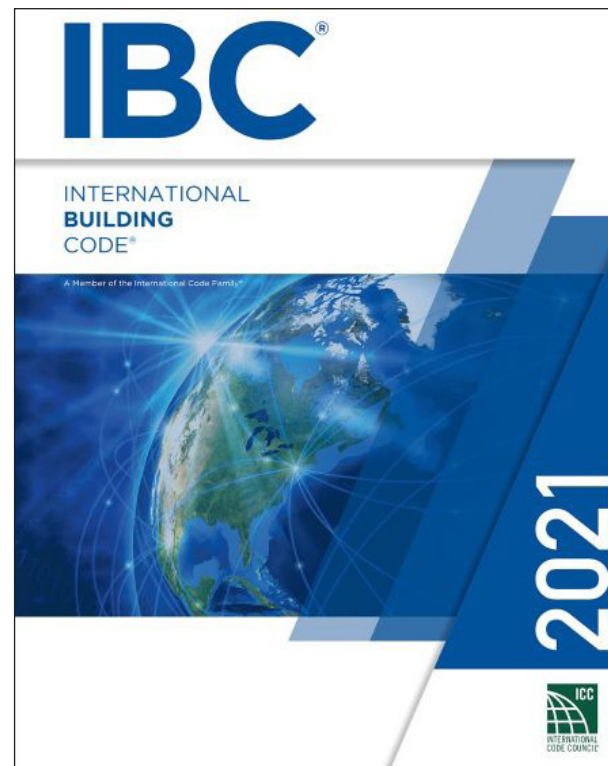
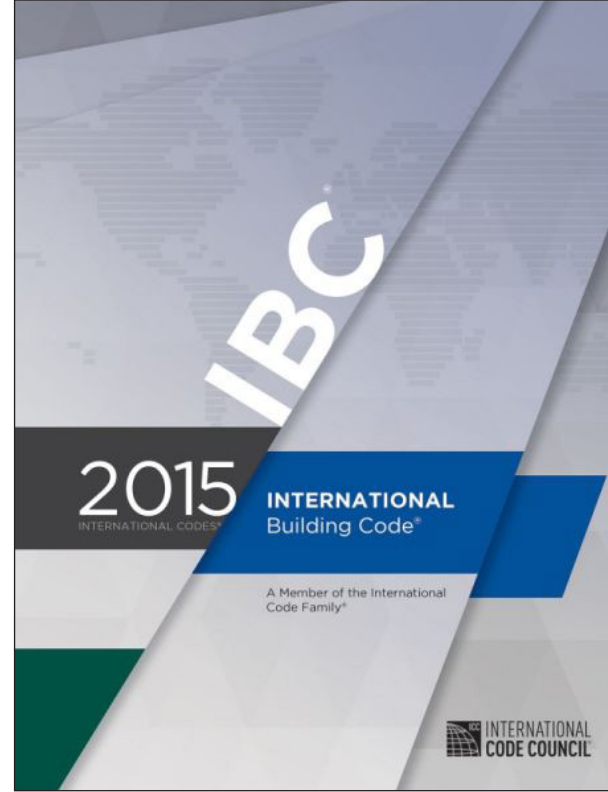
- The IBC is the primary building code in the United States, adopted (often with local amendments) by all 50 states, Washington, DC, and US territories
- The IBC is updated every 3 years
- Individual states, counties, and cities determine when and how to adopt a new edition of the code, resulting in regional variations and lag between publication and implementation
- It is essential to confirm which edition of the International Building Code (IBC) is in-use in a project's jurisdiction, as this determines how mass timber can be used in a project

MASS TIMBER IN THE IBC

- 2000: The first edition of the IBC included Type IV: Heavy Timber
- 2015: Cross-Laminated Timber (CLT) was explicitly introduced as a building material in the IBC, marking the first formal inclusion of mass timber
- 2021: Three new building types were introduced: Type IV-A, Type IV-B, and Type IV-C allowing mass timber to be used in buildings up to 18 stories tall
- 2024: Further code advancements expanded allowable exposed timber, particularly within Type IV-B

FIRE-RESISTANCE RATING (FRR)

The 2024 IBC defines **Fire-Resistance Rating (FRR)** as "the period of time a building element, component or assembly maintains the ability to confine a fire, continues to perform a given structural function, or both, as determined by the tests, or the methods based on tests, prescribed in Section 703." (For further reading, see Section 703: Fire-Resistance Ratings and Fire Tests).



The 2015, 2018, 2021, and 2024 editions of the IBC

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Basic Design Criteria: Construction Type & Fire-Resistance Rating

2018 IBC TABLE 601:

FIRE-RESISTANCE RATING REQUIREMENTS FOR BUILDING ELEMENTS (HOURS)

Construction Type		Non-Combustible				Combustible Allowed				
		TYPE I		TYPE II		TYPE III		TYPE IV	TYPE V	
Building Element		A	B	A	B	A	B	HT	A	B
Primary Structural Frame		3	2	1	0	1	0	HT	1	0
Bearing Walls	Exterior	3	2	1	0	2	2	2	1	0
	Interior	3	2	1	0	1	0	1/HT	1	0
Non-Bearing Walls	Exterior	See Table 602								
	Interior	0	0	0	0	0	0	See Section 2304.11.2	0	0
Floor		2	2	1	0	1	0	HT	1	0
Roof		1.5	1	1	0	1	0	HT	1	0

2018 IBC

Table 601 from the 2018 IBC lists the fire-resistance rating requirements for five construction types:

- Type I: Noncombustible, high fire resistance
- Type II: Noncombustible, low fire resistance
- Type III: Noncombustible exterior, combustible interior
- Type IV: Noncombustible exterior, heavy timber interior
- Type V: Combustible

Combustible materials are allowed for the primary structural frame in Type III, Type IV, and Type V, meaning timber construction is allowed. These construction types are shaded in red.

Type IV does not have an explicit fire rating, only “HT” (heavy timber), which limits timber member sizes to empirically established dimensions as listed in Table 2304.11: Minimum Dimensions of Heavy Timber Structural Members. The oversizing of these heavy timber structural members corresponds to approximately a 1-hour fire resistance rating based on empirical evidence.

Basic Design Criteria: Construction Type & Fire-Resistance Rating

2021/2024 IBC TABLE 601:
FIRE-RESISTANCE RATING REQUIREMENTS FOR BUILDING ELEMENTS (HOURS)

Construction Type		Non-Combustible				Combustible Allowed							
		TYPE I		TYPE II		TYPE III		TYPE IV			TYPE V		
Building Element		A	B	A	B	A	B	A	B	C	HT	A	B
Primary Structural Frame		3	2	1	0	1	0	3	2	2	HT	1	0
Bearing Walls	Exterior	3	2	1	0	2	2	3	2	2	2	1	0
	Interior	3	2	1	0	1	0	3	2	2	1/HT	1	0
Non-Bearing Walls	Exterior					See Table 705.5							
	Interior	0	0	0	0	0	0	0	0	0	See Section 2304.11.2	0	0
Floor		2	2	1	0	1	0	2	2	2	HT	1	0
Roof		1.5	1	1	0	1	0	1.5	1	1	HT	1	0

2021 IBC

The 2021 IBC introduced three new construction types to account for mass timber in tall buildings.

- Type IV-A: High fire resistance, no exposed mass timber allowed
- Type IV-B: Moderate fire resistance, exposed mass timber allowed within prescribed surface area limits
- Type IV-C: Moderate fire resistance, exposed mass timber allowed

Note that the FRR for Type IV-A is identical to Type I-A, and Types IV-B and IV-C are identical to Type I-B. These conservative limits were necessary to get approval for such a significant code change.

While Type IV-A allows for the tallest mass timber building construction, Type IV-C allows for the greatest use of exposed mass timber structure.

2024 IBC

The 2024 IBC did not introduce any new construction types or updates to the FRR established in the 2021 IBC, but increased exposed mass timber allowances in Type IV-B, and refined how exposed surface area is calculated.

Basic Design Criteria: Construction Type & Fire-Resistance Rating

CONSTRUCTION TYPE (SPRINKLERED VALUES - S, SM)

Occupancies	I-A	I-B	III-A	IV-A	IV-B	IV-C	IV-HT
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IBC TABLE 504.3: ALLOWABLE BUILDING HEIGHT IN FEET ABOVE GRADE PLANE (S)

A, B, R	Unlimited	180	85	270	180	85	85
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IBC TABLE 504.4: ALLOWABLE NUMBER OF STORIES ABOVE GRADE PLANE (S)

A-2, A-3, A-4	Unlimited	12	4	18	12	6	4
B	Unlimited	12	6	18	12	9	6
R-2	Unlimited	12	5	18	12	8	5

IBC TABLE 506.2: ALLOWABLE AREA FACTOR (A_T = SM) IN SQUARE FEET

A-2, A-3, A-4	Unlimited	Unlimited	42,000	135,000	90,000	56,250	45,000
B	Unlimited	Unlimited	85,500	324,000	216,000	135,000	108,000
R-2	Unlimited	Unlimited	72,000	184,500	123,000	76,875	61,500

Selected data from 2024 IBC

FEASIBILITY OF A MASS TIMBER BUILDING

The most critical step in designing a mass timber building is to ensure that mass timber is allowed for a given project’s construction type and occupancy group, as well as its intended height and area. At a minimum, there must be a clear path toward approval for a configuration that does not meet prescriptive code limits.

COMMON MASS TIMBER BUILDING SIZE LIMITS

The maximum size of a mass timber building is affected by both its construction type and occupancy group, affecting allowable height, number of stories, and area.

The chart at the left shows comparative data for occupancy groups A (Assembly), B (Business), and R (Residential). In mass timber construction, allowable floor area is frequently the limiting factor, though height and number of stories are also constraints.

Basic Design Criteria: Tall Wood Buildings



Ascent, Milwaukee, WI, 2022; Architect: Korb Architecture, Structural Engineer: Thorton Thomasetti

TRENDS IN TALL WOOD CONSTRUCTION

As cities continue to densify, there is a growing interest in tall wood buildings. These new buildings can offer both environmentally-responsible material selections and biophilic environments for workplaces, housing, and public spaces.

The 2021 IBC's three new construction types not only increased allowable building heights, but also redefined how architects and structural engineers understand the relationship between building height, fire-resistance ratings, and allowable timber exposure in mass timber construction.

In general, taller mass timber buildings require higher FRRs and increased encapsulation, resulting in less visible wood. Architects prioritizing the visual expression of timber often explore hybrid mass timber solutions, such as limiting mass timber to one portion of a high rise, or the pursuit of alternative means and methods through code provisions that allow for performance-based solutions.

Today, tall timber is a global phenomenon, with new construction across North America, Europe, and Oceania each year. As of 2025, Ascent (Milwaukee, WI) is the tallest hybrid mass timber building in the world.

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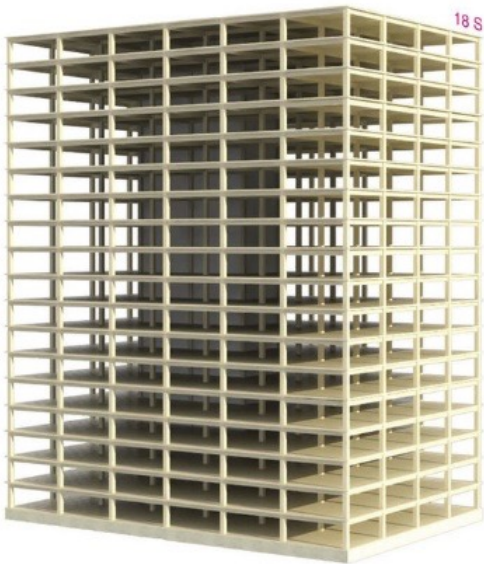
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Basic Design Criteria: Tall Wood Buildings

TALL WOOD & CONSTRUCTION TYPES

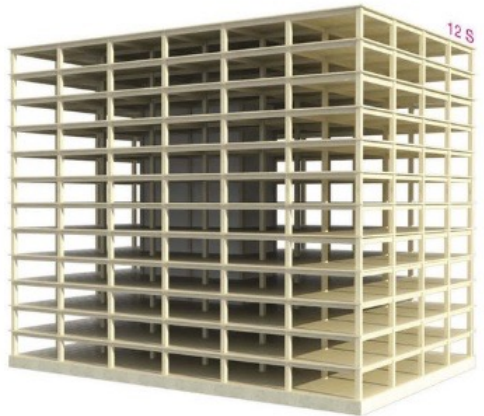
In mass timber projects, a clear understanding of a client’s program needs is paramount to establish the occupancy and construction type early in the design process. These parameters determine the allowable number of stories, height, area, and extent of timber exposure, which in turn inform both conceptual design and early cost estimates.

Exposed timber relies on the principle that wood burns at a slow and predictable rate, allowing a sacrificial outer layer of wood to form a char layer that protects the inner wood fibers. Meeting required FRRs therefore requires oversizing structural members, which directly links fire performance, member sizing, and construction type.



TYPE IV-A

18 STORIES	
BUILDING HEIGHT	270 FT
ALLOWABLE AREA	972,000 SF
AVERAGE AREA/STORY	54,000 SF
PRIMARY FRAME FRR	3-HR
WOOD EXPOSURE	NONE



TYPE IV-B

12 STORIES	
BUILDING HEIGHT	180 FT
ALLOWABLE AREA	648,000 SF
AVERAGE AREA/STORY	54,000 SF
PRIMARY FRAME FRR	2-HR
WOOD EXPOSURE	20% CEILING 40% WALLS



TYPE IV-C

9 STORIES	
BUILDING HEIGHT	85 FT
ALLOWABLE AREA	405,000 SF
AVERAGE AREA/STORY	45,000 SF
PRIMARY FRAME FRR	2-HR
WOOD EXPOSURE	ALL



TYPE IV-HT

6 STORIES	
BUILDING HEIGHT	85 FT
ALLOWABLE AREA	324,000 SF
AVERAGE AREA/STORY	54,000 SF
PRIMARY FRAME FRR	PRESCRIPTIVE
WOOD EXPOSURE	ALL

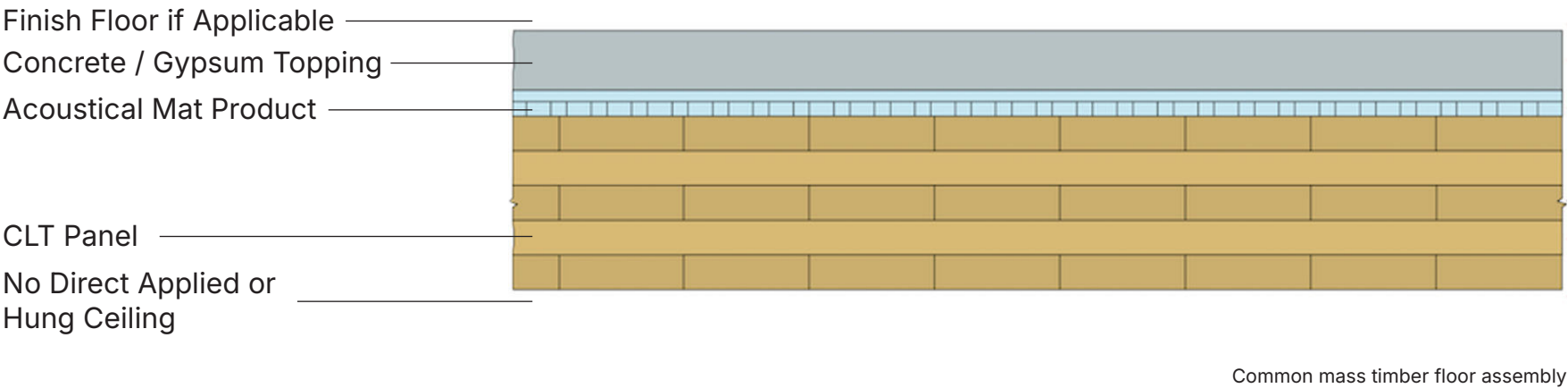
Comparative study of height limits and FRR for business occupancy

Design for Performance: Acoustics

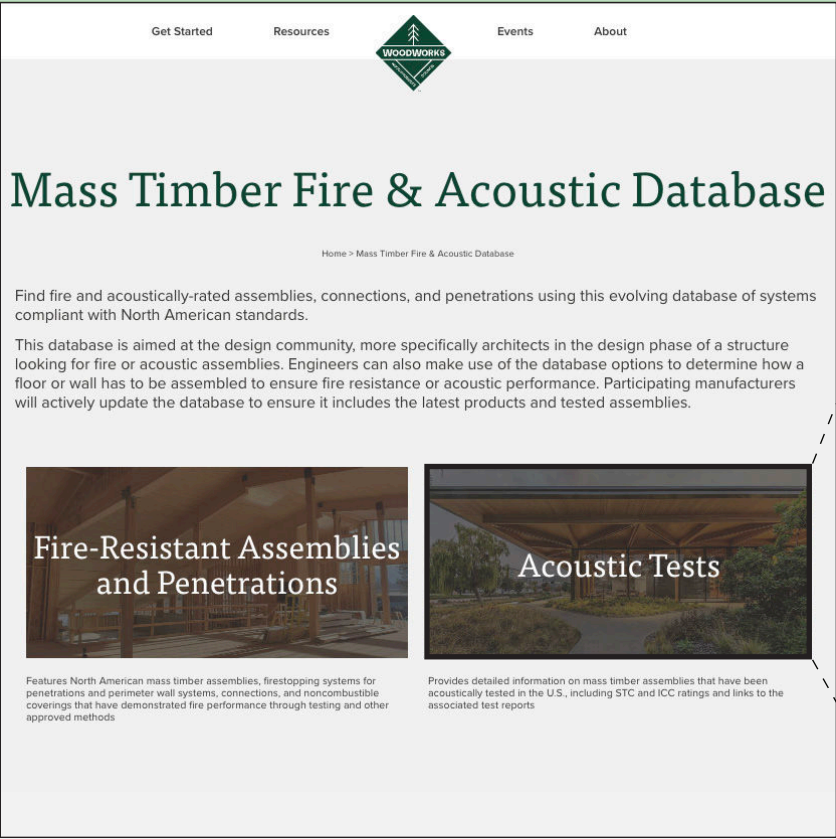
Mass timber is light and structurally efficient compared to concrete. This is generally a benefit, reducing gravity loads and therefore foundation demands and the overall volume of concrete in a mass timber building. However, sound transmission and vibration control are mass-driven, making mass timber assemblies more sensitive to sound transmission and vibration.

As a result, achieving high acoustic performance may be a challenge for acoustically sensitive programs such as residential, hospitality, healthcare, and offices, where privacy and occupant comfort are a priority. For example, IBC Section 1206.2: Airborne-borne sound requires a minimum Sound Transmission Class (STC) rating of 50, or a field-tested Normalized Noise Isolation Class (NNIC) of at least 45, between dwelling units, sleeping units, and adjacent public areas.

As with many mass timber design challenges, acoustical performance can be effectively managed with early coordination. The components of an acoustic assembly may have a significant effect on the floor depth, which in turn can affect floor-to-ceiling height, building height, and cost.



Design for Performance: Acoustics



WoodWorks website

Mass Timber Panel	Acoustical Product Between MT Panel and Topping	Topping	Finish Floor	Sound Rating	Impact Rating	Method of Compliance
5-layer 6.88" CLT	Maxxon Acousti-Mat® SBR over Maxxon Acousti-Mat® 3/4 Premium	2" Gyp-Crete®	None	58 STC ①	49 IIC ①	Maxxon / Intertek Report # P4428.01-113-11-R0 Contact Product Manufacturer for Additional Information
5-layer 6.88" CLT	Maxxon Acousti-Mat® SBR over Maxxon Acousti-Mat® 3/4 Premium	2" Gyp-Crete®	LVT on Shaw GroundWorks	57 STC ①	54 IIC ①	Maxxon / Intertek Report # P4428.01-113-11-R0 Contact Product Manufacturer for Additional Information
5-layer 6.00" CLT	Maxxon Acousti-Mat® 3/4 Premium	2" Gyp-Crete®	Click LVT	52 STC ①	47 IIC ①	Veneklassen Associates / Williams + Paddon Architects + Planners Inc. Report # FC22-0392 Contact Product Manufacturer for Additional Information
5-layer 6.00" CLT	Maxxon Acousti-Mat® 3/4 Premium	2" Gyp-Crete®	Glue Down LVT	53 STC ①	47 IIC ①	Veneklassen Associates / Williams + Paddon Architects + Planners Inc. Report # FC22-0393 Contact Product Manufacturer for Additional Information
5.00" CLT(SCL)-MPP	Maxxon Acousti-Mat® 3/4 Premium	1.5" Gyp-Crete®	LVT on Acousti-Top®	53 STC ①	44 IIC ①	Maxxon / Intertek Report # K1283.01-113-11-R0 Contact Product Manufacturer for Additional Information
5.00" CLT(SCL)-MPP	Maxxon Acousti-Mat® SBR over Maxxon Acousti-Mat® 3/4	2" Concrete	None	57 STC ①	47 IIC ①	Maxxon / Intertek Report # K1282.01-113-11-R0 Contact Product Manufacturer for Additional Information
5.00" CLT(SCL)-MPP	Maxxon Acousti-Mat® SBR over Maxxon Acousti-Mat® 3/4	2" Concrete	LVT on Acousti-Top®	57 STC ①	51 IIC ①	Maxxon / Intertek Report # K1282.02-113-11-R0 Contact Product Manufacturer for Additional Information
6.37" CLT(SCL)-VLT	Maxxon Acousti-Mat® 3/8 Premium	2" Gyp-Crete®	None	53 STC ①	41 IIC ①	Maxxon / Intertek Report # L3827.03-113-11-R0 Contact Product Manufacturer for Additional Information
6.37" CLT(SCL)-VLT	Maxxon Acousti-Mat® 3/8 Premium	2" Gyp-Crete®	LVT on Milliken Premium	53 STC ①	50 IIC ①	Maxxon / Intertek Report # P2528.02-113-11-R0 Contact Product Manufacturer for Additional Information
6.37" CLT(SCL)-VLT	Maxxon Acousti-Mat® SBR over Maxxon Acousti-Mat® 3/4 Premium	2" Gyp-Crete®	None	59 STC ①	49 IIC ①	Maxxon / Intertek Report # L3827.55-113-11-R0 Contact Product Manufacturer for Additional Information
6.37" CLT(SCL)-VLT	Maxxon Acousti-Mat® SBR over Maxxon Acousti-Mat® 3/4 Premium	2" Gyp-Crete®	LVT	58 STC ①	52 IIC ①	Maxxon / Intertek Report # L3827.56-113-11-R0 Contact Product Manufacturer for Additional Information
6.37" CLT(SCL)-VLT	Maxxon Acousti-Mat® SBR over Maxxon Acousti-Mat® 3/4 Premium	2" Gyp-Crete®	LVT on Acousti-Top®	57 STC ①	54 IIC ①	Maxxon / Intertek Report # L3827.57-113-11-R0 Contact Product Manufacturer for Additional Information
5-layer 6.88" CLT	Maxxon Acousti-Mat® 3/8 Premium	2" Gyp-Crete®	None	55 STC ①	47 IIC ①	Maxxon / Intertek Report # Q7356.01-113-11-R0 Contact Product Manufacturer for Additional Information
5-layer 6.88" CLT	Maxxon Acousti-Mat® 3/8 Premium	2" Gyp-Crete®	LVT on 2mm Armstrong Pad	55 STC ①	47 IIC ①	Maxxon / Intertek Report # G2463.08-113-11-R0 Contact Product Manufacturer for Additional Information
6.37" CLT(SCL)-VLT	Maxxon Acousti-Mat® 3/8 Premium	2" Gyp-Crete®	Engineered Wood	52 STC ①	48 IIC ①	Maxxon / Intertek Report # L3827.08-113-11-R0 Contact Product Manufacturer for Additional Information

Excerpt from WoodWorks' acoustic database

FREE RESOURCES

WoodWorks offers many robust, free design resources, including a regularly updated database of acoustically tested mass timber assemblies. The database helps architects and engineers quickly identify floor and wall systems to meet owner and code requirements.

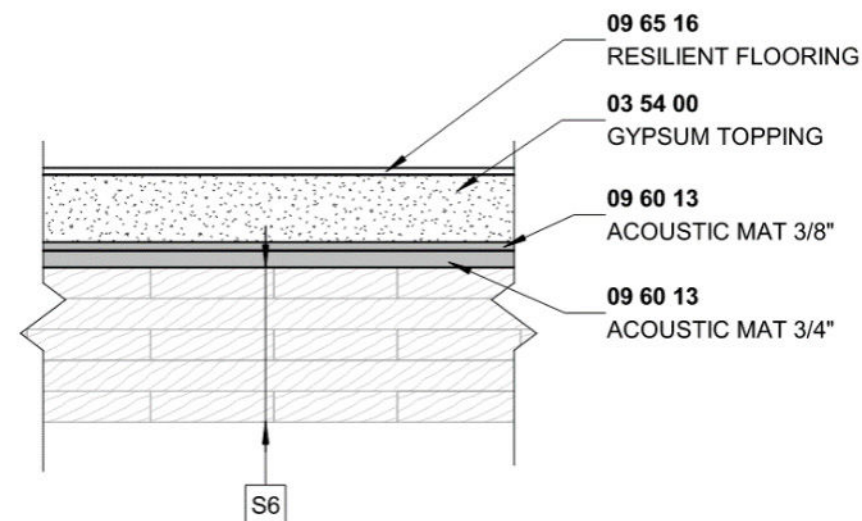
Design for Performance: Acoustics

HIGHER ACOUSTICAL PERFORMANCE

High-end residential and hospitality clients often target STC and IIC ratings of 60-65, exceeding code minimums. Achieving this level of acoustic performance with mass timber assemblies can be a challenge; designers should be aware that achieving high STC and IIC ratings may require greater total floor assembly depth.

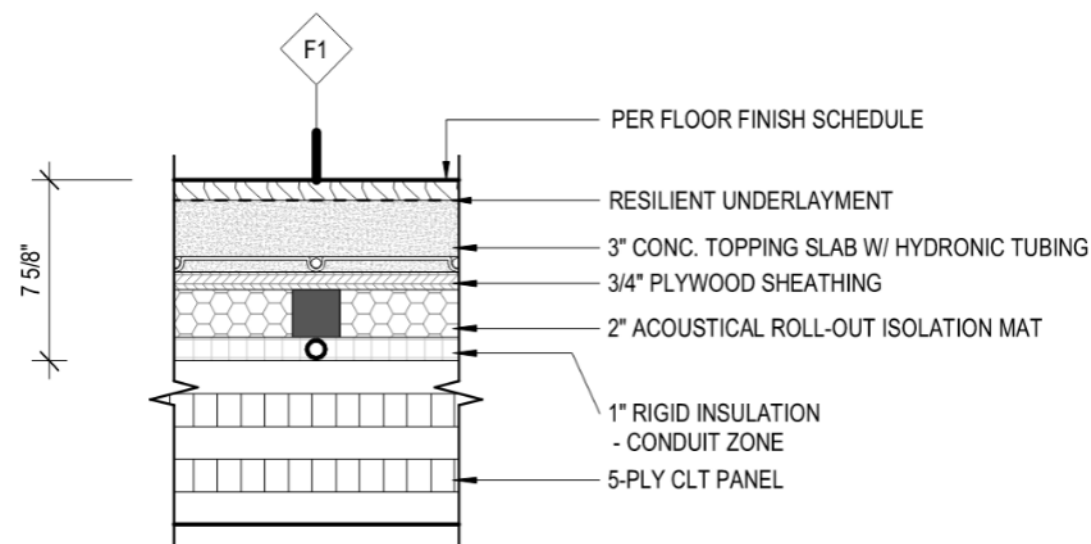
Acoustical engineers also call attention to flanking sounds, where sound travels around an assembly rather than through it. In mass timber assemblies, continuous panels and structural connections can create flanking paths that limit achievable acoustic performance. Research and testing in this area is ongoing.

CODE MINIMUM



Example floor assembly meeting code minimum for acoustical performance in approximately 4" thick buildup (in addition to timber structure) for STC 50

HIGHER PERFORMANCE



Example floor assembly achieving higher acoustical performance in approximately 7-5/8" buildup (in addition to timber structure) for STC 58

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Design for Performance: MEP Systems



MEP systems obscuring view of mass timber structure

MEP SYSTEM DESIGN

In typical building projects, mechanical, electrical, and plumbing (MEP) system design is often deferred to a design-build contractor late in the design process. For mass timber projects, MEP systems are yet another area that benefits from early coordination.

- MEP routing strategies influence the structural grid, column layout, and floor-to-floor height, with direct implications for the building program and cost
- Some mass timber construction types have specific restrictions or requirements for concealed spaces through which fire can spread, making late-stage concealment of MEP systems difficult or costly
- Timber is less tolerant of unplanned field-cut penetrations in structural members than steel or concrete. Poorly coordinated openings can compromise structural capacity and fire performance
- The visual expression of wood is a primary design goal in many mass timber buildings. Without early coordination, MEP systems may impede this goal, resulting in ducts, pipes, and cable trays that visually dominate or obscure the timber structure

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Structural Design with Mass Timber

Design for Performance: MEP Strategies



The Loading Dock, Boulder, CO, 2017;
Architect: OZ Architecture; Structural Engineer: KL&A Engineers and Builders

1. Minimizing Floor Assembly Depth

This approach minimizes the structural depth and routes larger MEP systems below the structure, with limited prefabricated penetrations through structural members for small diameter services.



T3, Minneapolis, MN, 2016; Architect: Michael Green Architecture,
Structural Engineer: Magnusson Klemencic Associates

2. Minimizing Selected Structural Depth

In the example above, a column line is located close to the core, minimizing the transverse beam depth in that bay. This allows MEP mains to run parallel to the core without constraining the head-height clearance.



Platte Fifteen, Denver, CO, 2020;
Architect: OZ Architecture; Structural Engineer: KL&A Engineers and Builders

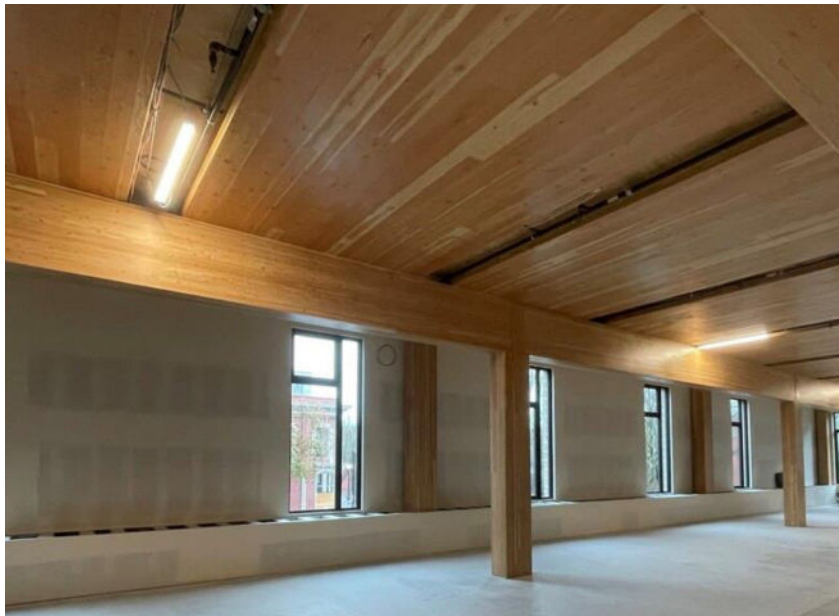
3. Layering Structure

Here, beams are set on top of girders, creating a 16-inch dedicated service zone for MEP routing. Larger ducts (not pictured) are relegated to back of house spaces, where possible, and electrical conduit is routed through the non-structural concrete topping. The strategy is cost effective and visually appealing, but increases the overall floor-to-floor depth.

**STRUCTURAL
MODULE 2**

Structural
Design with
Mass Timber

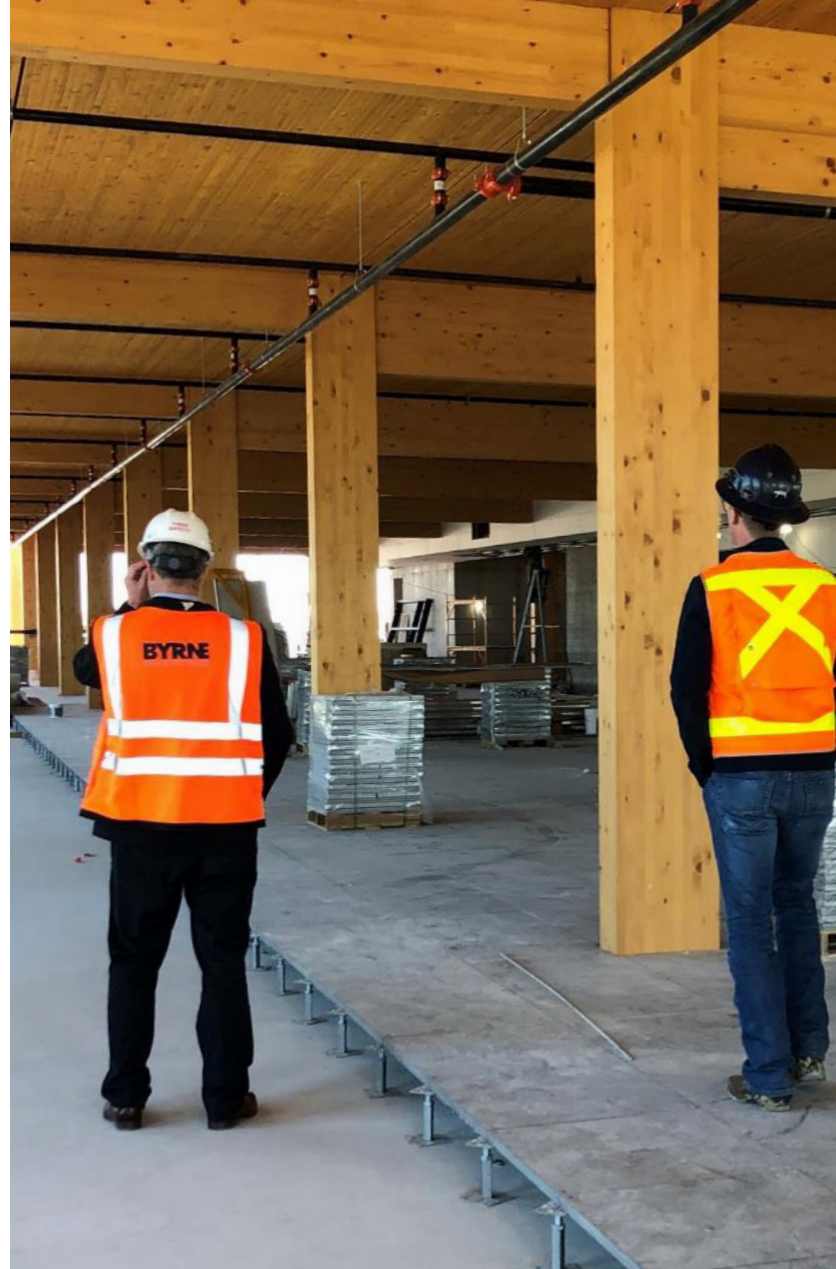
Design for Performance: MEP Strategies



The PAE Living Building, Portland, OR, 2021;
Architect: ZGF Architects, Structural Engineer: KPFF Consulting Engineers

4. Sprinklered Concealed Space

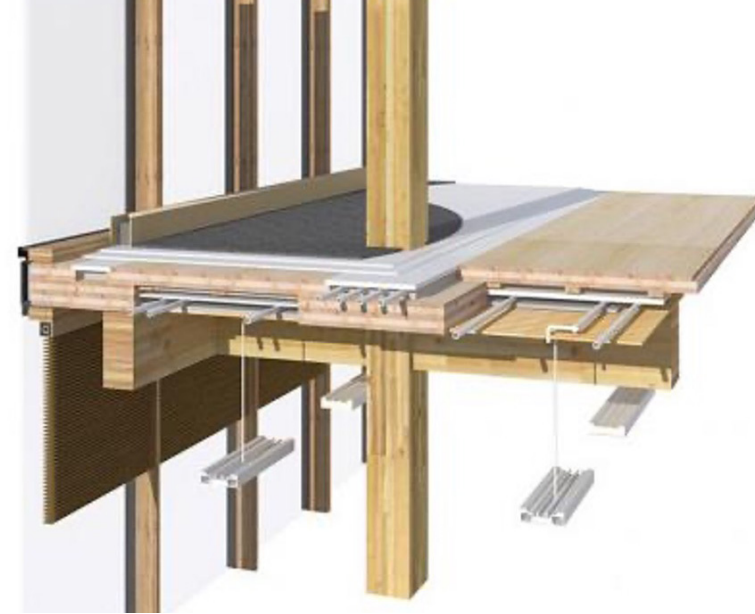
To maintain a flush timber ceiling, a sprinklered concealed space is located between floor panels. In this example, the structural floor topping allows the cavity to remain serviceable without compromising the diaphragm capacity of the floor.



The Soto, San Antonio, TX, 2020;
Architect: Lake|Flato Architects, BOKA Powell, Structural Engineer: StructureCraft

5. Raised Floor

A fire-protected raised floor accommodates most MEP services without exposed ducts below the timber ceiling.



Wood Innovation and Design Centre, Prince George, BC, Canada, 2014;
Architect: Michael Green Architecture; Structural Engineer: Equilibrium Consulting

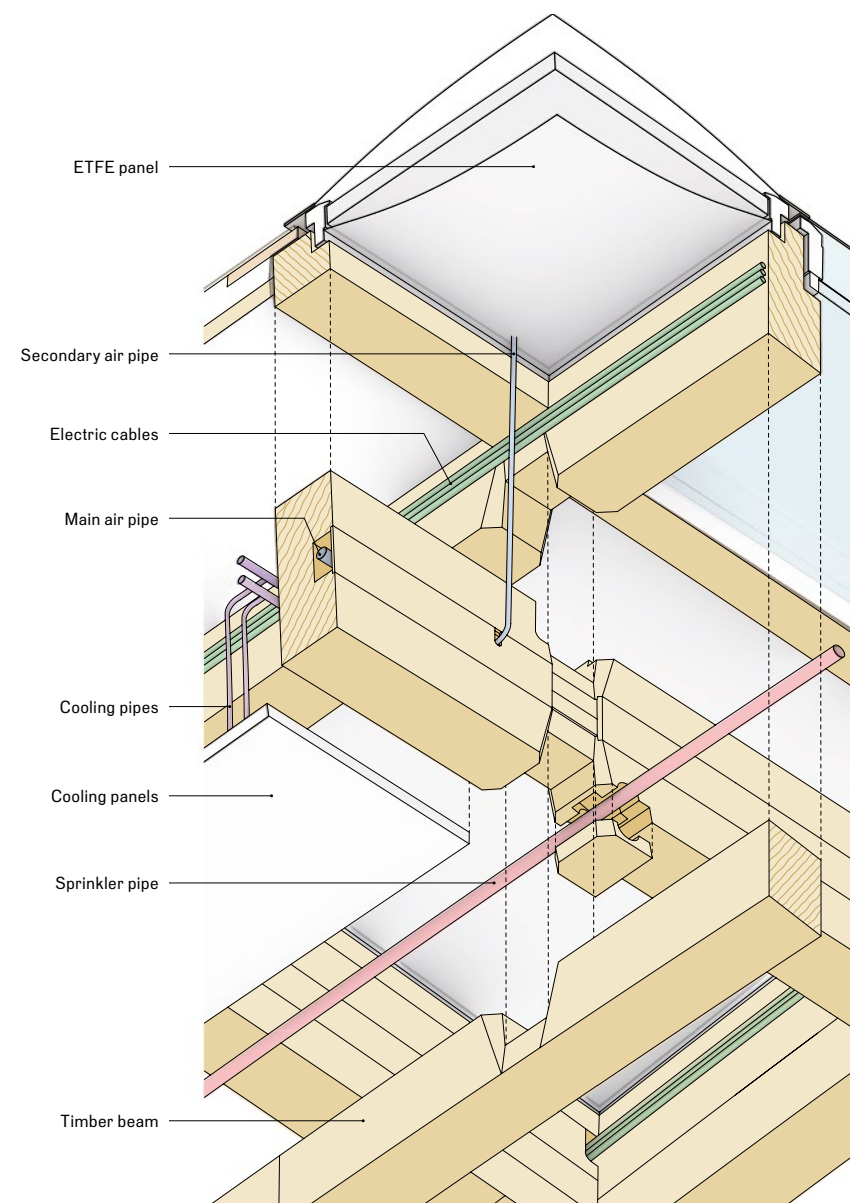
6. Two-Layer Panel System

This system uses gaps within a two-layer panel system to route mechanical and electrical systems between panels both above and below the wood floor plane.

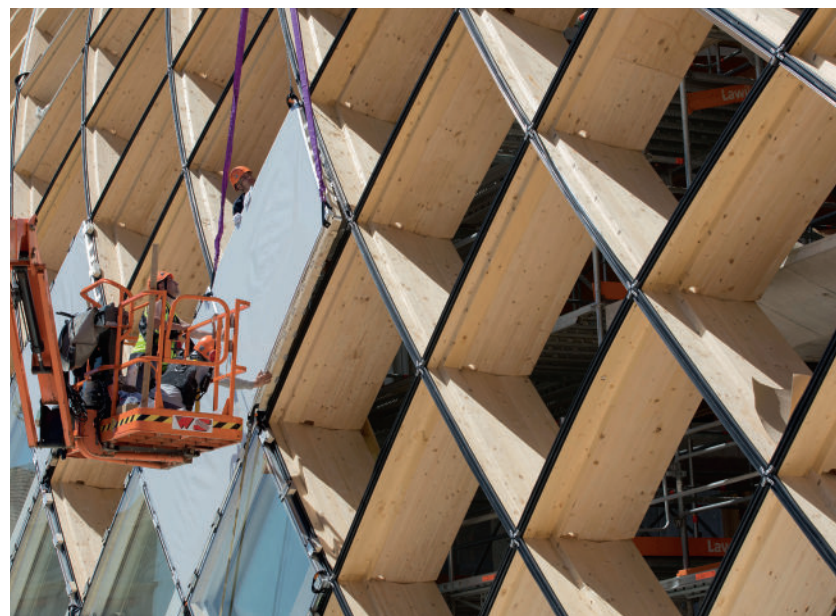
STRUCTURAL MODULE 2

Structural
Design with
Mass Timber

Design for Performance: MEP Strategies



Swatch Headquarters, Biel/Bienne, Switzerland, 2019; Architect: Shigeru Ban Architects, Structural Engineer: Cr ation Holz



7. Integrated Building Envelope

Integrating the facade, structure, and building services into a single coordinated assembly rather than discrete layers minimizes envelope depth, but requires a high level of coordination throughout project phases.

At the Swatch Headquarters, this approach was developed using a detailed 3D digital model in place of traditional construction documents. The model was adapted for fabrication of all timber elements, minimizing construction errors.

Prefabricated openings in the timber structure were precisely sized for electrical cables, sprinkler pipes, air pipes, and cooling systems, with cutouts carefully staggered and oriented to maintain structural integrity and account for the direction of the wood grain.

Though the building systems remain exposed in the finished space, their visual presence does not compete with the timber. By integrating the facade, structure, and MEP systems, this project reduced its envelope depth by 33%, preserving usable floor area.

STRUCTURAL MODULE 2

Structural
Design with
Mass Timber

Design for Economy: Commodity Lumber Pricing

In mass timber construction, the first cost of the structural system can be a challenge. However, this does not have to be a prohibitive hurdle: there are design strategies that directly impact structural economy, such as the structural grid, member proportions, and material selection.

Commodity lumber prices — the cost of wood per unit of volume — are largely beyond the control of architects and engineers. While some manufacturers are vertically integrated, with ownership of forests, mills, and fabrication facilities, most rely on external commodity markets for raw materials. As a result, the cost of fabricated mass timber products often rises and falls with the price of logs or lumber.

Commodity prices are shaped by a wide range of forces, from natural phenomena such as weather and insect infestations, to geopolitical factors including wars, sanctions, and trade barriers. The graph at the right shows market fluctuations from January 2016 – December 2025. Late 2017 through early 2019 were once considered an era of profound variability, with an unprecedented twofold magnitude difference in cost per board foot in less than one year. However, this volatility pales in comparison to the COVID pandemic's impact on lumber prices from 2020–2022.



**STRUCTURAL
MODULE 2**

**Structural
Design with
Mass Timber**

Design for Economy: Volume of Timber

COMMON CLT LAYUPS: VOLUME RELATIVE TO THIN 3-PLY

Volume Relative to Thin 3-Ply	Layup Composition, Thickness	Cross-Section	Layup Composition, Thickness	Cross-Section
1.0 - 1.2x	3-ply 3-layer			
	(3.43" - 4.14") (87 mm - 105 mm)			
1.6 - 2.0x	5-ply 5-layer			
	(5.47" - 6.9") (139 mm - 175 mm)			
2.2 - 2.8x	7-ply 7-layer		7-ply 5-layer	
	(7.52" - 9.66") (191 mm - 245 mm)		(7.52" - 9.66") (191 mm - 245 mm)	
2.8 - 3.6x	9-ply 9-layer		9-ply 7-layer	
	(9.57" - 12.42") (243 mm - 315 mm)		(9.57" - 12.42") (243 mm - 315 mm)	

Comparative wood volumes of CLT layups

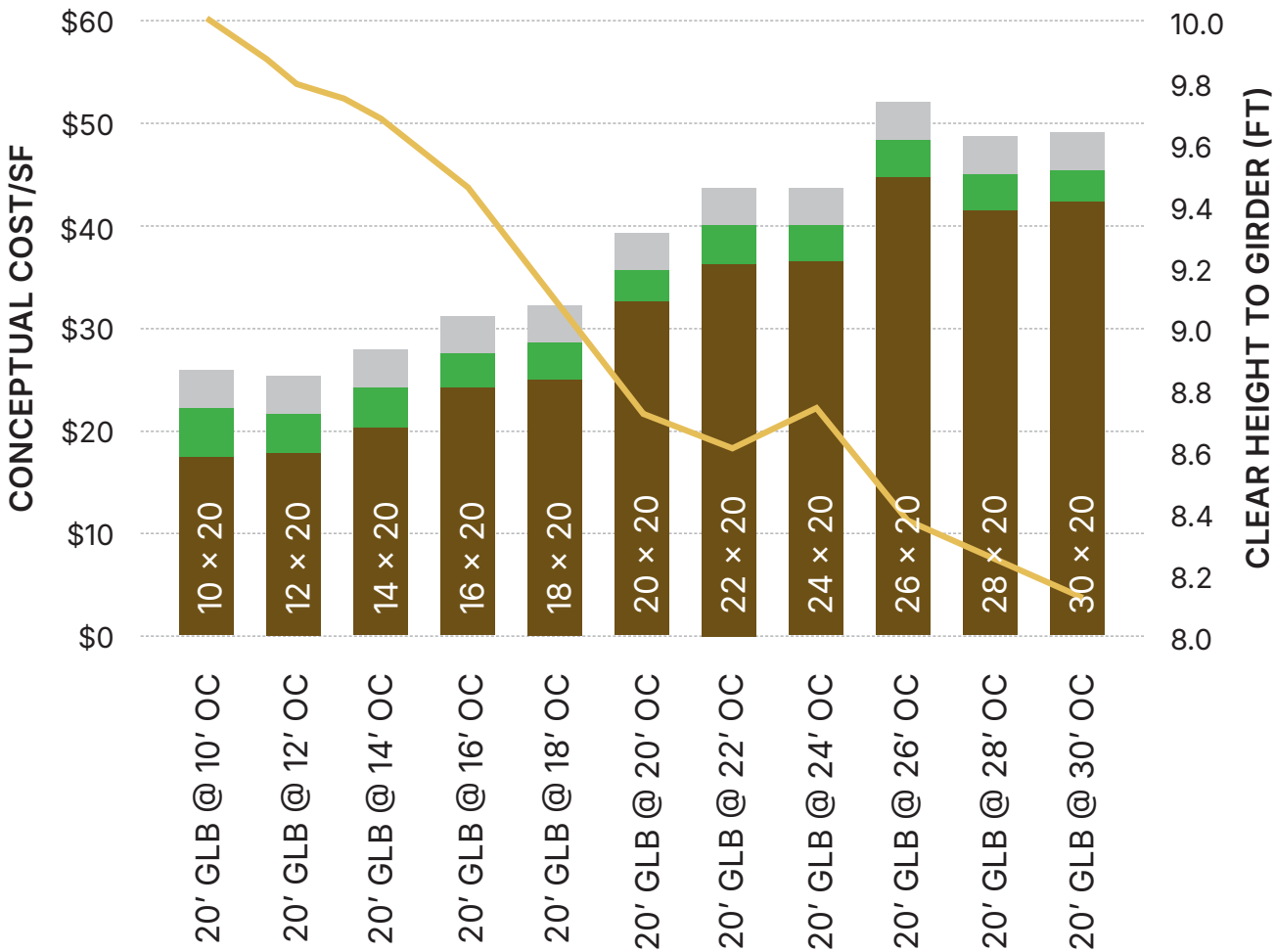
In a hierarchy of economic drivers, the second most important factor is the total volume of timber in a project. Architects and structural engineers work together to select material systems and member sizing to meet program, structural, and code requirements as economically as possible.

The diagram at the left shows common CLT layups, showing the range of available panel thicknesses, as well as the relative volume of wood for each panel system as compared to a standard 3-ply panel.

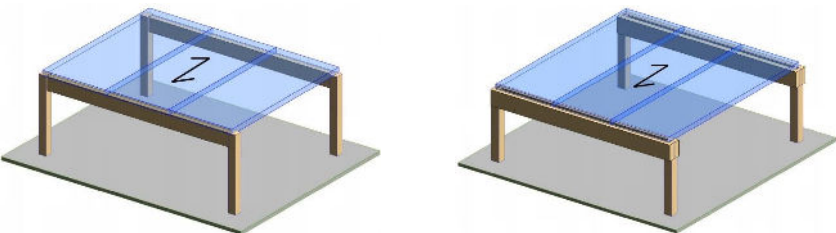
Mass timber cost is not perfectly linear with timber volume, but it is closely correlated. A design primarily utilizing 7-ply CLT panels may require 2.5x as much wood as a 3-ply system, resulting in a substantial increase in cost — potentially 2.5x as much. For this reason, most mass timber projects are designed to use 3-ply or 5-ply CLT panels. 7-ply is uncommon, and 9-ply is rare in practice.

Design for Economy: Optimizing the Structural Grid

CASE 1: 20' GIRDER, NO BEAMS



- CONCRETE TOPPING
- WOOD CONNECTIONS
- WOOD
- CLEAR HEIGHT



CLT can span long distances, but as the thickness increases, so does the volume of wood fiber, and thus the cost. But reducing the structural grid spacing is not feasible for all programs.

STRUCTURAL MODULE COMPARISON

The following slides present a comparative study of structural modules for a hypothetical system with CLT floors and glulam girders. Cost data is intended for relative comparison only, not absolute pricing.

- X-axis: Glulam girder spacing, on-center
- Y-axis (left): Cost per square foot
- Vertical bars: Total structural cost
 - Gray: concrete topping
 - Green: wood connections
 - Brown: wood
- Y-axis (right) and yellow line: Clear height from floor to underside of timber structure

CASE 1

- 20' glulam girders
- No beams
- CLT spans between girders

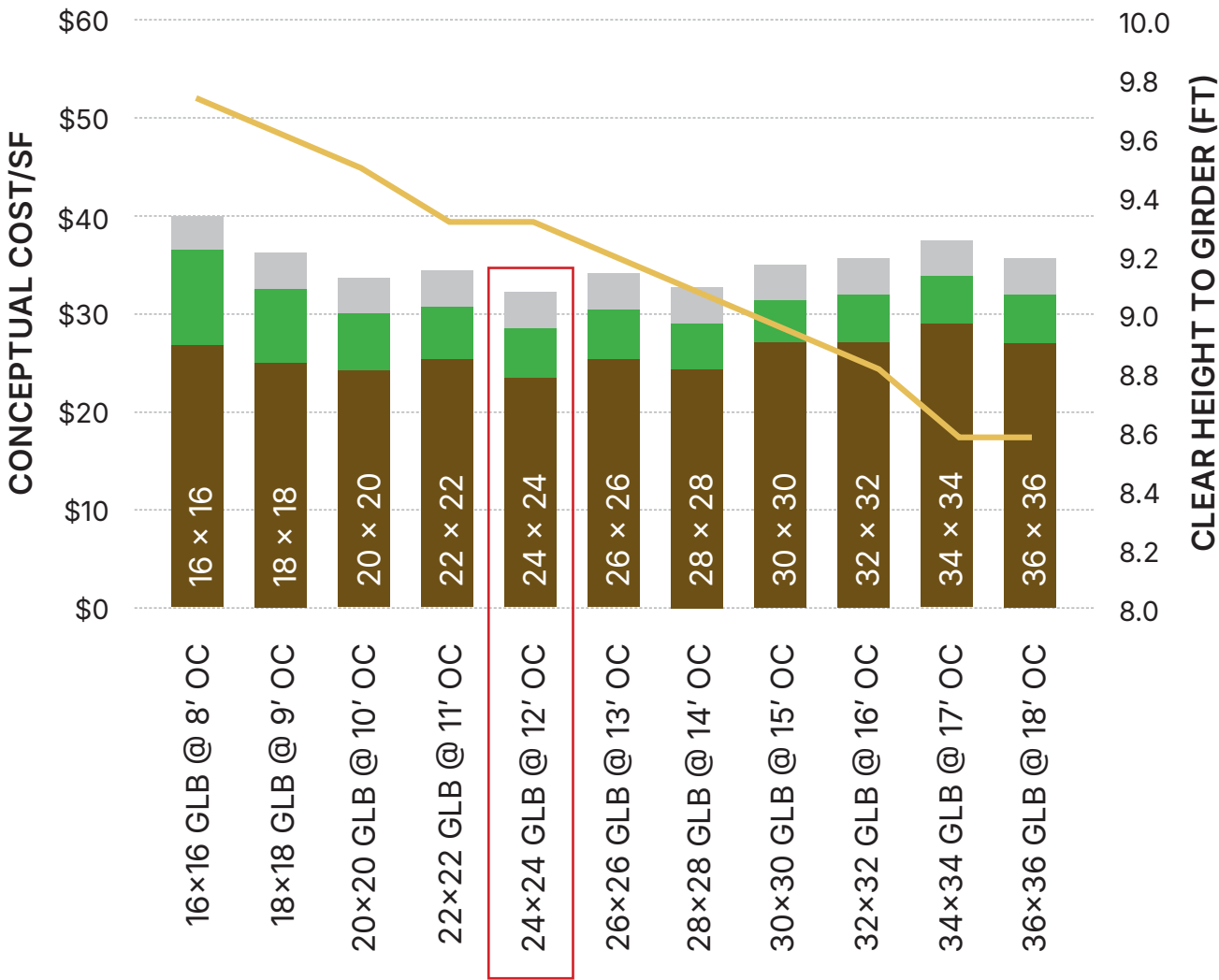
Case 1 demonstrates a largely linear relationship between increased span and increased cost.

STRUCTURAL MODULE 2

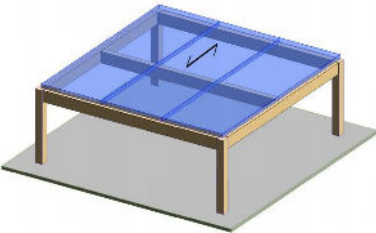
Structural Design with Mass Timber

Design for Economy: Optimizing the Structural Grid

CASE 2: SQUARE BAY WITH SECONDARY BEAM



- CONCRETE TOPPING
- WOOD CONNECTIONS
- WOOD
- CLEAR HEIGHT



CASE 2

- Square bay of varying length
- One beam centered in bay
- CLT spans equally in two directions

Case 2 introduces a midspan beam while maintaining square structural proportions. In this configuration, the optimal grid spacing is 24', highlighted in red. Larger structural grids show a steep loss in clear height due to deeper beams and girders.

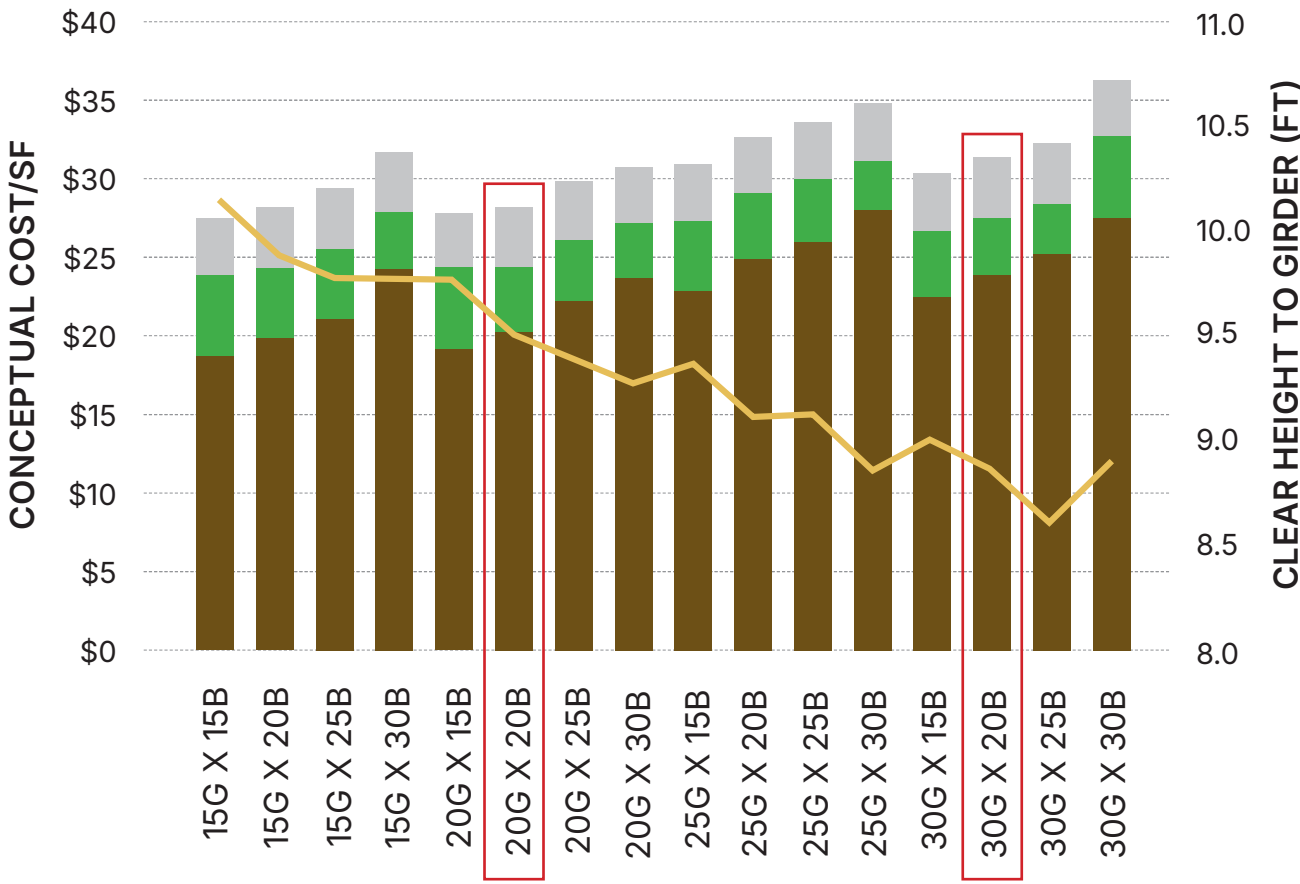
Initially, the cost per square foot of connections decreases as bay size increases because fewer connections are distributed over a larger area. At larger spans, however, connection cost per square foot increases as the structural demand on each connection increases.

STRUCTURAL
MODULE 2

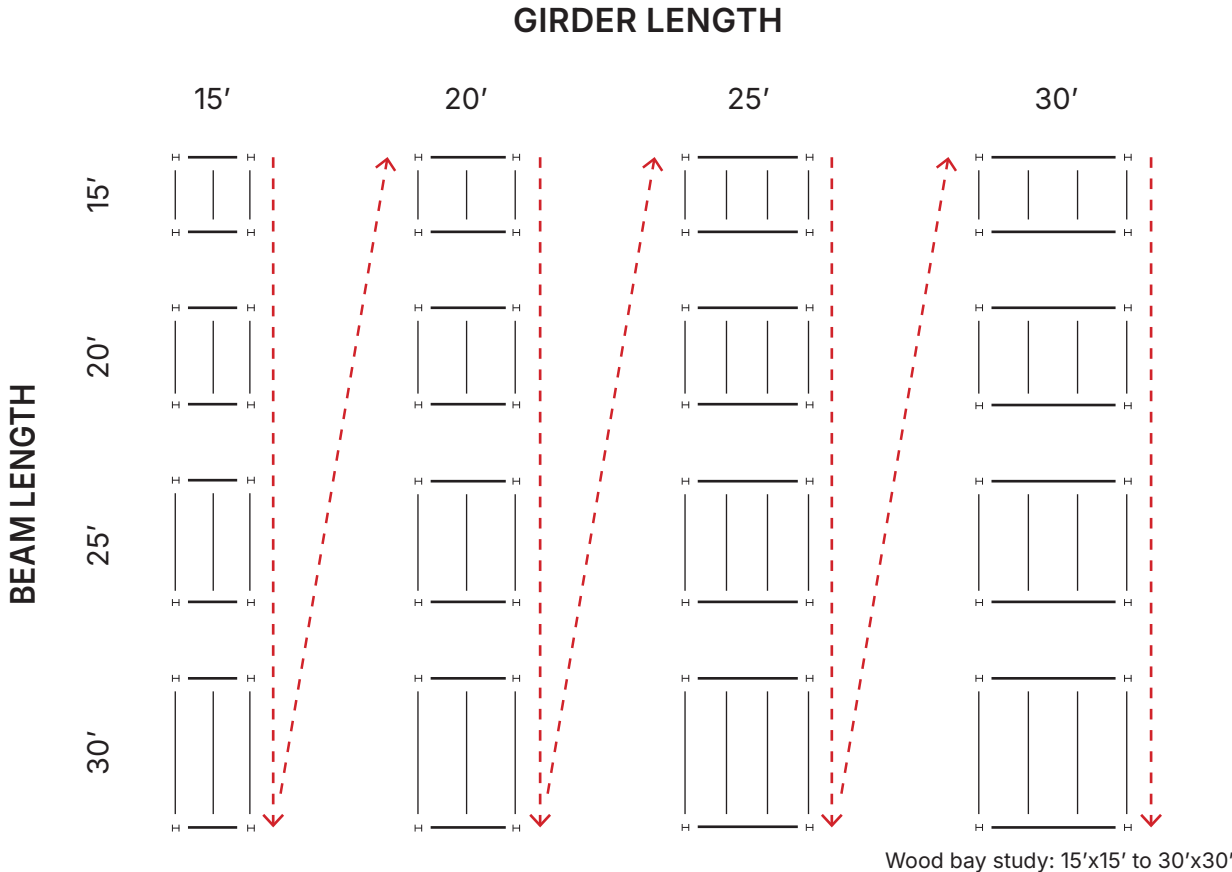
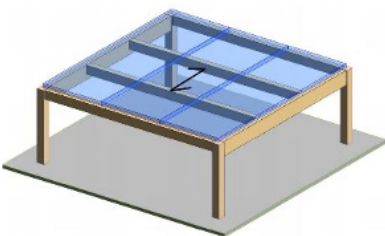
Structural
Design with
Mass Timber

Design for Economy: Optimizing the Structural Grid

CASE 3: 3-PLY CLT ON TIMBER GRID
5' GRID INCREASES FROM 15'x15' TO 30'x30' WITH SECONDARY BEAMS



CONCRETE TOPPING
WOOD CONNECTIONS
WOOD
CLEAR HEIGHT



CASE 3

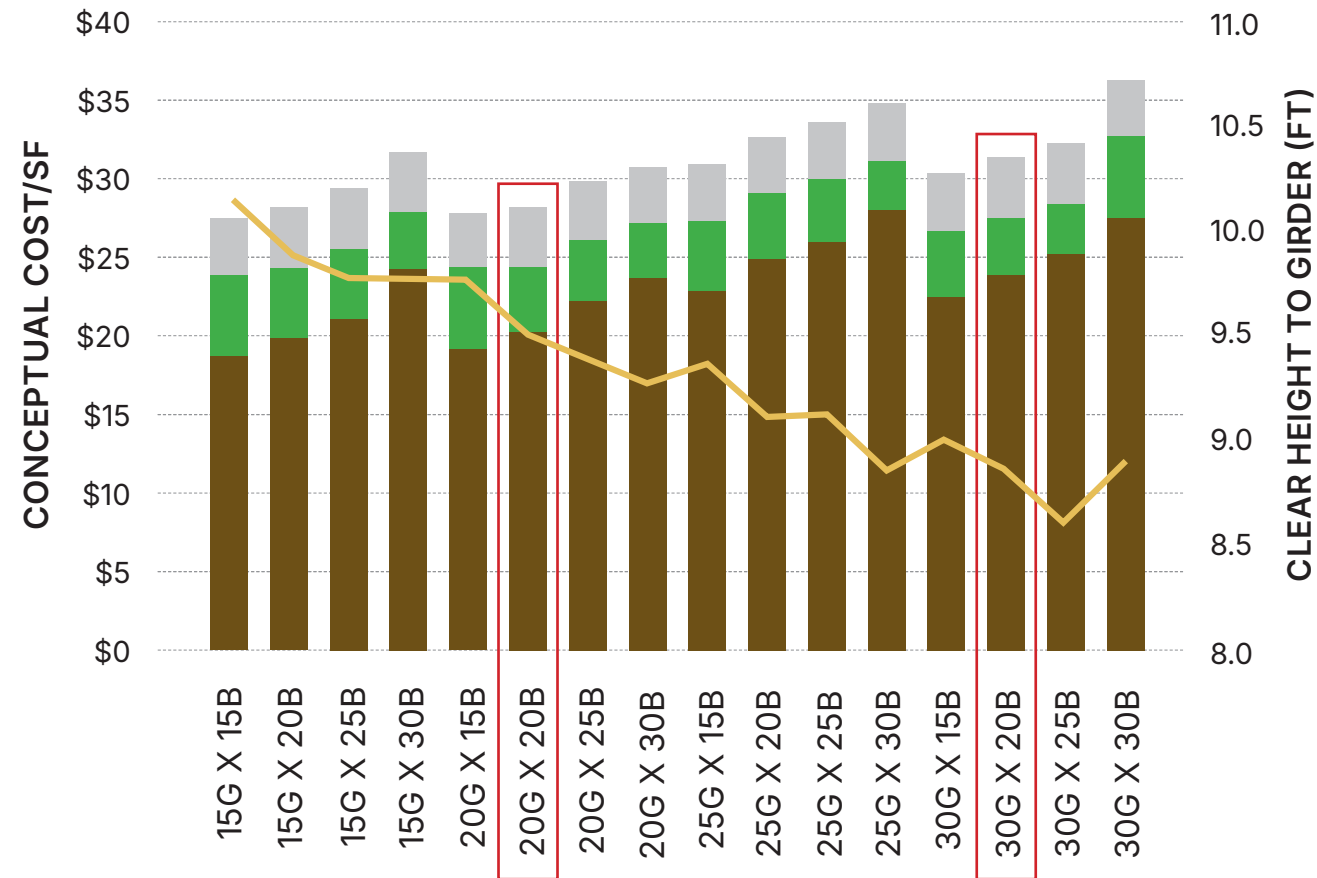
- Comparative study with 5' increases in structural grid
- Beam spans increased first, followed by girder spans
- Beams centered or evenly spaced
- CLT spans equally between supports

Cost increases in a stair-step pattern, with optimal grid spacing at smaller structural grids with shorter beam spans, where girder and beam depth, and therefore timber volume, are minimized.

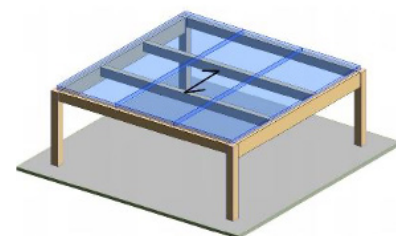
Design for Economy: Optimizing the Structural Grid

CASE 3: 3-PLY CLT ON TIMBER GRID

5' GRID INCREASES FROM 15'×15' TO 30'×30' WITH SECONDARY BEAMS



CONCRETE TOPPING
WOOD CONNECTIONS
WOOD
CLEAR HEIGHT

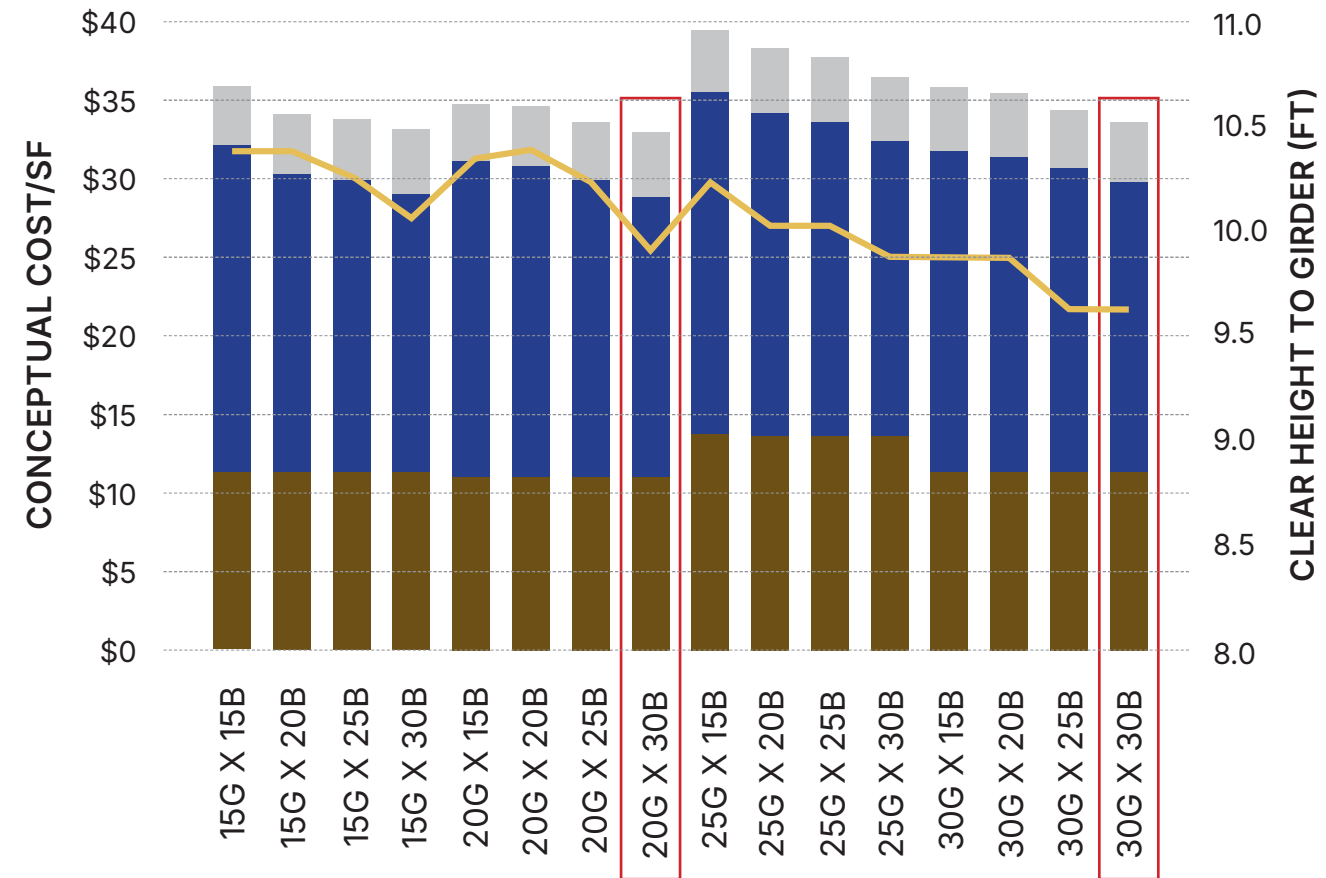


< **Timber Systems**
Material volume dominates cost

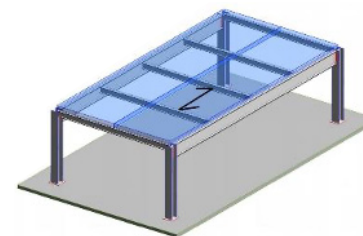
Steel Systems
Labor dominates cost >

CASE 4: HYBRID – 3-PLY CLT ON STEEL GRID

5' GRID INCREASES FROM 15'×15' TO 30'×30' WITH SECONDARY BEAMS



CONCRETE TOPPING
STEEL
WOOD
CLEAR HEIGHT



CASE 4

- Identical grid study as Case 3, but with steel girders, beams, and columns
- Beams centered or spaced equidistant
- CLT with equal spans (varies)

In contrast to Case 3, costs decrease in a stair-step pattern, with optimal grid spacing at larger grids with longer spans and fewer connections. An economical mass timber building cannot be built using the same structural grids that have produced economical steel buildings.

**STRUCTURAL
MODULE 2**

Structural
Design with
Mass Timber

Design for Economy: DfMA Across Construction Types

While grid optimization can meaningfully affect cost within a construction type, the selection of construction type itself has a greater impact due to differences in FRR, which result in drastic differences in timber volume, structural member depth, and the need for noncombustible protection.

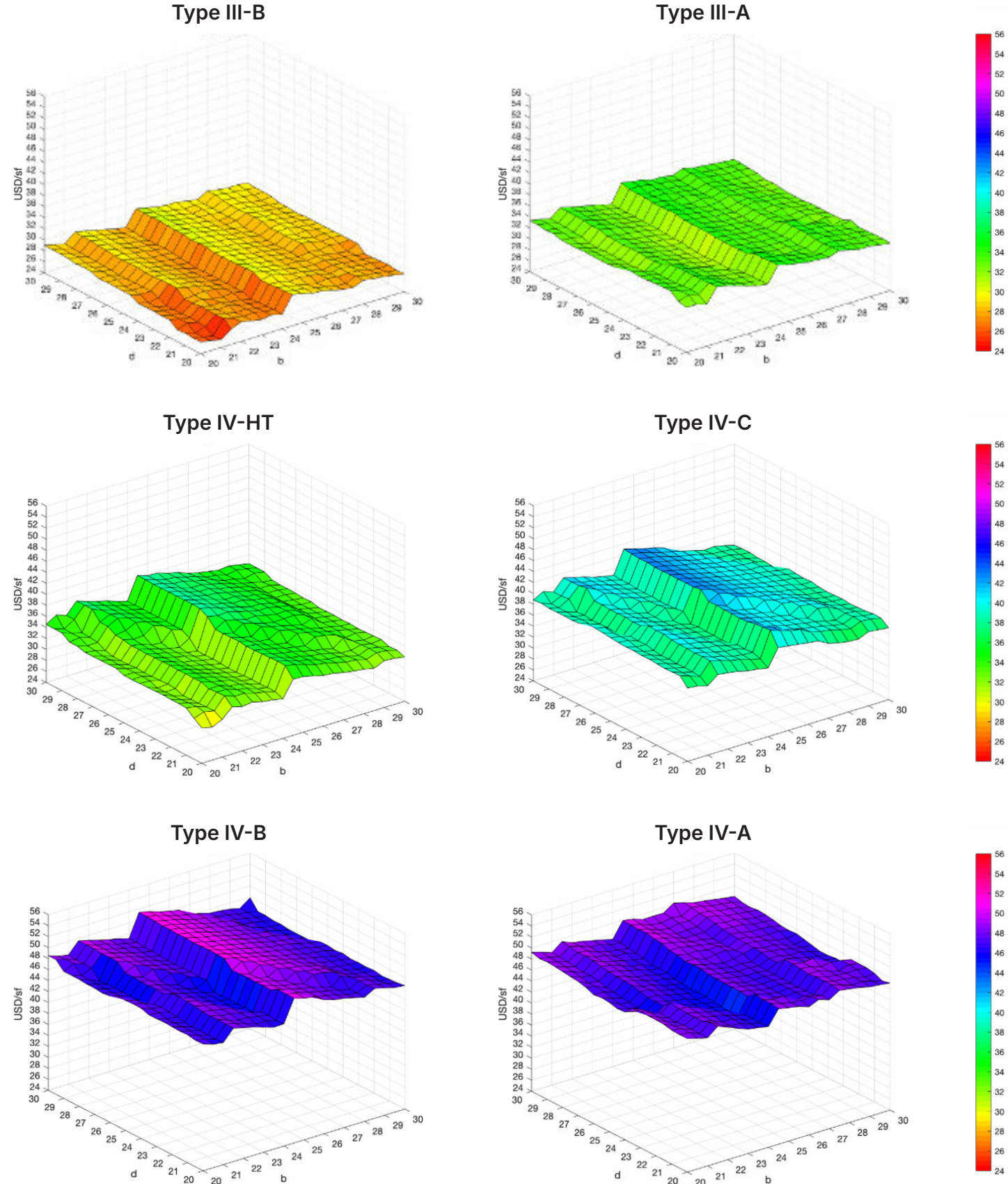
The six adjacent surface plots compare hundreds of structural grid configurations across multiple construction types.

- X- and Y-axes: structural bay dimensions
- Z-axis (color scale): structural cost per square foot

Within a construction type, the surface plot reveals opportunities for Design for Manufacturing and Assembly (DfMA) optimization:

- Step-like ridges reflect manufacturing constraints, such as CLT panel press lengths relative to project cut lengths
- Diagonal ridges reflect increases in connection costs as structural demands increase

DfMA can help optimize cost within a construction type, but the selection of construction type and the associated FRR has the greatest impact on overall structural cost.



Comparative surface plots of construction types, structural bay dimensions, and unit costs

**STRUCTURAL
MODULE 2**

**Structural
Design with
Mass Timber**

Design for Constructibility: Supply Chain



Selection of timber manufacturers across the United States

In a commodified material system, such as wide-flange steel sections, an architect or engineer can reasonably assume that products from different suppliers are functionally equivalent.

Mass timber is not yet commodified, meaning that it is not standardized to the point of interchangeability. While standards such as ANSI/APA PRG 320 Standard for Performance-Rated Cross-Laminated Timber exist, the market is characterized by evolving supply chains and meaningful variations among manufacturers, both domestically and internationally.

Today, every mass timber project is effectively a custom design: unlike purchasing lumber off the shelf, mass timber panel orders are scheduled, fabricated, and often engineered specifically for a given project.

Engaging manufacturers and fabricators as early as possible is a clear benefit, both informing the design process and supporting budget and schedule goals.

**STRUCTURAL
MODULE 2**

**Structural
Design with
Mass Timber**

Design for Constructibility: Supply Chain



Hydraulic press for manufacturing CLT



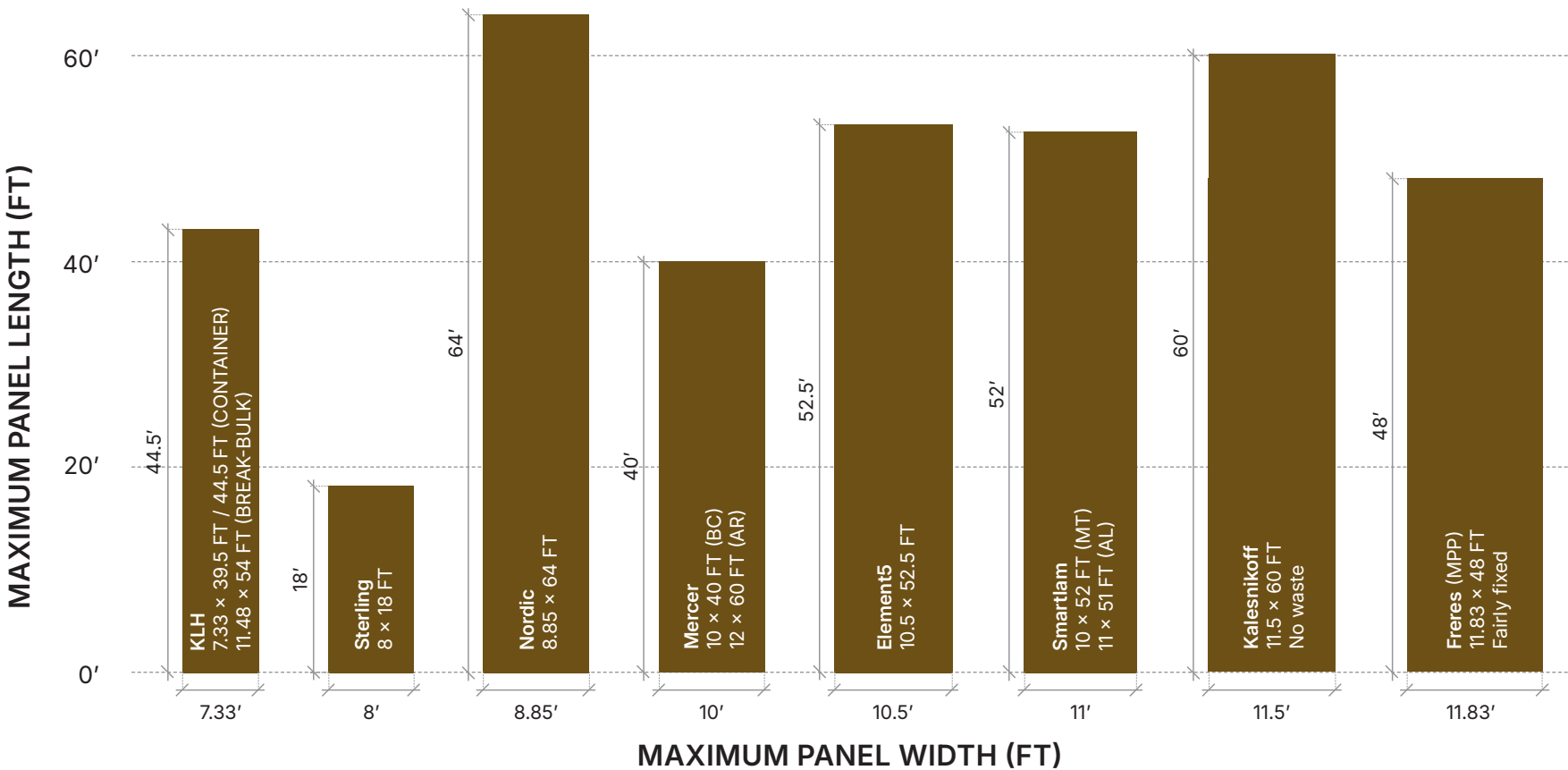
Wood samples by species. Left to right: Spruce-Pine-Fir (SPF) & Spruce-Pine-Fir South (SPF-S), Douglas Fir-Larch (DFL), Hem-Fir (HF), Southern Yellow Pine (SYP)

Each manufacturer has unique offerings in:

- Panel width
- Panel length
- Laminated thickness
- E-rated (machine stress rated)
- V-rated (visually graded)
- Species
- Special finishes

Designing without awareness of manufacturers' offerings and capabilities can lead to increased costs and late-stage redesign.

CLT AND MPP PANEL SIZES BY MANUFACTURER



Design for Constructibility: Manufacturing Standards

AMERICAN NATIONAL STANDARD

ANSI/APA PRG 320-2025

Standard for Performance-Rated Cross-Laminated Timber







Manufacturing standard for CLT in the United States

ANSI/APA PRG 320 offers generic member properties, which can be sufficient at the very beginning of a project. But different suppliers offer a wide variety of species, grades, and configurations that are not captured in these material tables.

ASD REFERENCE DESIGN VALUES ^a FOR BASIC CLT GRADES AND LAYUPS (FOR USE IN THE U.S.)																	
CLT Grade	t _b (in.)	Lamination Thickness (in.) in CLT Layup								Major Strength Direction				Minor Strength Direction			
		=	⊥	=	⊥	=	⊥	=		(F _b S) _{eff,1.0} (lb _f -ft/ ft of width)	(EI) _{eff,1.0} (10 ⁶ lb _f - in. ² /ft of width)	(GA) _{eff,1.0} (10 ⁶ lb _f / ft of width)	V _{5.0} (lb _f /ft of width)	(F _b S) _{eff,1.90} (lb _f -ft/ft of width)	(EI) _{eff,1.90} (10 ⁶ lb _f - in. ² /ft of width)	(GA) _{eff,1.90} (10 ⁶ lb _f /ft of width)	V _{5.90} (lb _f /ft of width)
E1	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8					4,525	115	0.46	1,490	160	3.1	0.61	495
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8			10,400	440	0.92	2,480	1,370	81	1.2	1,490
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	18,375	1,089	1.4	3,475	3,150	313	1.8	2,480
E2	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8					3,825	102	0.53	1,980	165	3.6	0.56	660
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8			8,825	389	1.1	3,300	1,440	95	1.1	1,980
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	15,600	963	1.6	4,625	3,300	364	1.7	3,300
E3	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8					2,800	81	0.35	1,160	110	2.3	0.44	385
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8			6,400	311	0.69	1,930	955	61	0.87	1,160
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	11,325	769	1.0	2,700	2,210	234	1.3	1,930
E4	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8					4,525	115	0.50	1,820	140	3.4	0.62	605
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8			10,400	440	1.0	3,025	1,230	88	1.2	1,820
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	18,400	1,089	1.5	4,225	2,850	338	1.9	3,025
E5	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8					3,825	101	0.46	1,650	160	3.1	0.55	550
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8			8,800	389	0.92	2,750	1,370	81	1.1	1,650
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	15,575	962	1.4	3,850	3,150	312	1.7	2,750
V1	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8					2,090	108	0.53	1,980	165	3.6	0.59	660
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8			4,800	415	1.1	3,300	1,440	95	1.2	1,980
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	8,500	1,027	1.6	4,625	3,300	364	1.8	3,300
V1[N]	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8					1,980	108	0.53	1,980	150	3.6	0.59	660
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8			4,550	415	1.1	3,300	1,300	95	1.2	1,980
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	8,025	1,027	1.6	4,625	3,000	364	1.8	3,300
V2	4 1/8	1 3/8	1 3/8	1 3/8	1 3/8					2,030	95	0.46	1,490	160	3.1	0.52	495
	6 7/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8			4,675	363	0.91	2,480	1,370	81	1.0	1,490
	9 5/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	1 3/8	8,275	898	1.4	3,475	3,150	312	1.6	2,480

Reference design values for basic CLT grades and layups

WoodWorks' Index of Mass Timber Connections is a helpful starting point for understanding the range of possible connection strategies. This resource catalogs over 100 connection typologies and documents key performance criteria, including:

- The breadth of the index underscores the inherent trade-offs in connection design between simplicity and flexibility, concealment and inspectability, and standardization and customization.

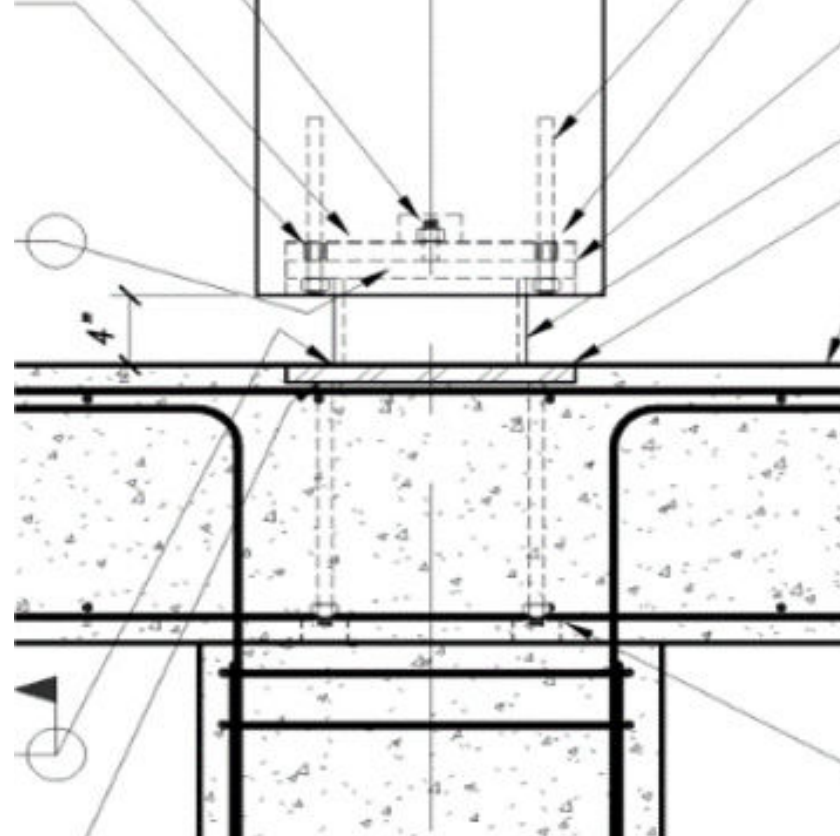
Connection design should also be considered in relation to design for deconstruction. It is important to consider the principles of circular economies, and how connections can enable future disassembly, material recovery, and reuse of mass timber components.



Design for Constructibility: Tolerances



Bearing plate



Detail drawing of column base

Project cost is closely tied to constructibility, which is often contingent on how effectively connections mediate between different structural materials. In concrete construction, the allowable tolerances are typically measured in inches, while mass timber adheres to a higher degree of precision, often measured in millimeters. Connections that fail to address this difference can slow construction as misalignments are reconciled in the field, which also increases labor costs.

The images at the left show a mass timber column supported by a concrete slab, detailed with tolerance in mind. The bearing plate is intentionally set low, then built up with washers and a secondary plate to correctly locate the elevation of the base of the timber itself. The space between the plates is then grouted to create a solid bearing surface before the timber column is installed.

Without this approach, variations in concrete elevation would likely have resulted in inconsistent column base heights across the floor, even if both the concrete and timber elements were fabricated and constructed within allowable tolerances. The strategy shown here anticipates the variability in concrete while maintaining the precise alignment required for the timber structure.



Bearing plate built up with washers and secondary plate



Timber column installed over column base

Design for Constructibility: Tolerances



Intended alignment of timber beam and concrete wall



Greater gap between timber beam and concrete wall than preferred

In mass timber construction, effective connection design often absorbs construction variability in a discreet manner, rather than demanding perfect alignment in the field.

The adjacent photos show two installations of the same timber-concrete connection. Although both installations meet design intent, variations in wall alignment result in visibly different gaps between the beam and the wall. The left photo shows the installation as intended, but rather than indicating poor construction, this pair of images is evidence of a well-designed connection that allows for out-of-plane variation while maintaining structural performance.

Conclusion

Mass timber buildings share many of the same design considerations as concrete, steel, and light-frame buildings, but reap even greater benefits from earlier and more integrated coordination across disciplines and supply chains.

This begins with a clear understanding of both the client and design team’s motivations for working with mass timber at the outset of a project. The team must also assess what is permitted by code, including construction type, fire-resistance rating, and limits on height, area, and wood exposure.

As a project advances, performance requirements such as acoustics and mechanical, electrical, and plumbing systems must be coordinated with structural decisions to ensure alignment with code requirements and project goals.

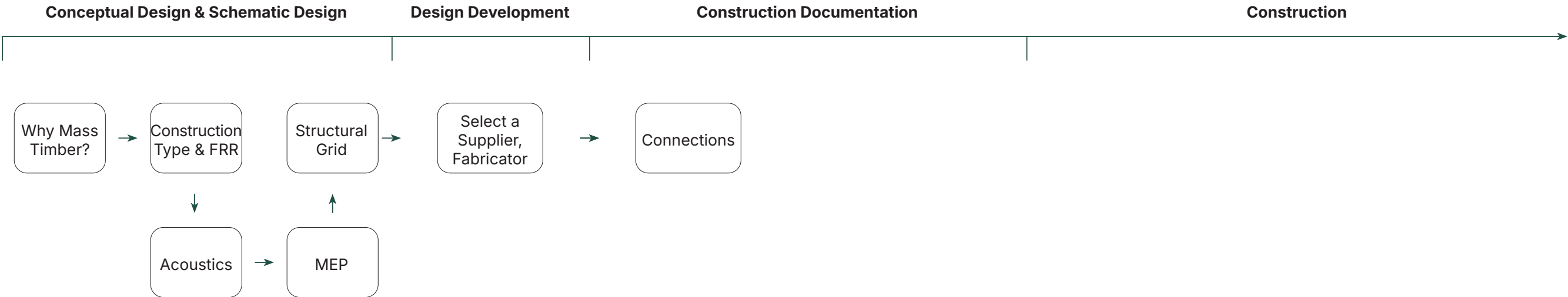
Although architects and engineers cannot control commodity lumber pricing, they can meaningfully influence cost through the timber volume, structural grid, and design for manufacturing and assembly. Constructibility depends on an early understanding of manufacturing standards, engagement with supply chains, and careful detailing of connections and tolerances.

Collectively, these factors underscore that the use of mass timber is not simply a material selection, but a shift in process, where early project decisions have an outsized impact on design, feasibility, cost, and overall project quality.

TRADITIONAL PROJECT SCHEDULE



MASS TIMBER PROJECT SCHEDULE



Citations

ENDNOTES

Page 07

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IMAGES, TABLES, AND GRAPHS

Page 01

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2. Shigeru Ban Architects. Photograph of Haesley Nine Bridges Golf Clubhouse. "Haesley Nine Bridges Golf Clubhouse." Accessed December 1, 2025. <https://shigerubanarchitects.com/wp-content/uploads/2023/06/4-2.jpg>

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3. Construction Users Roundtable. MacLeamy Curve diagram. *Collaboration, Integrated Information, and the Project Lifecycle in Building Design, Construction and Operation*, White Paper WP-1202 (August 2004), 16. Reformatted by authors.

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4. Kalesnikoff. Sawmill Factory, British Columbia, Canada. Accessed December 16, 2025. <https://kalesnikoff.com/>

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7. International Code Council. *International Building Code 2015*, Cover.

8. International Code Council. *International Building Code 2018*, Cover.

9. International Code Council. *International Building Code 2021*, Cover.

10. International Code Council. *International Building Code 2024*, Cover.

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